

# Notes for *Fundamentals of Microelectronics* (Razavi) (2nd edition, 2014)

## 《微电子基本原理》笔记

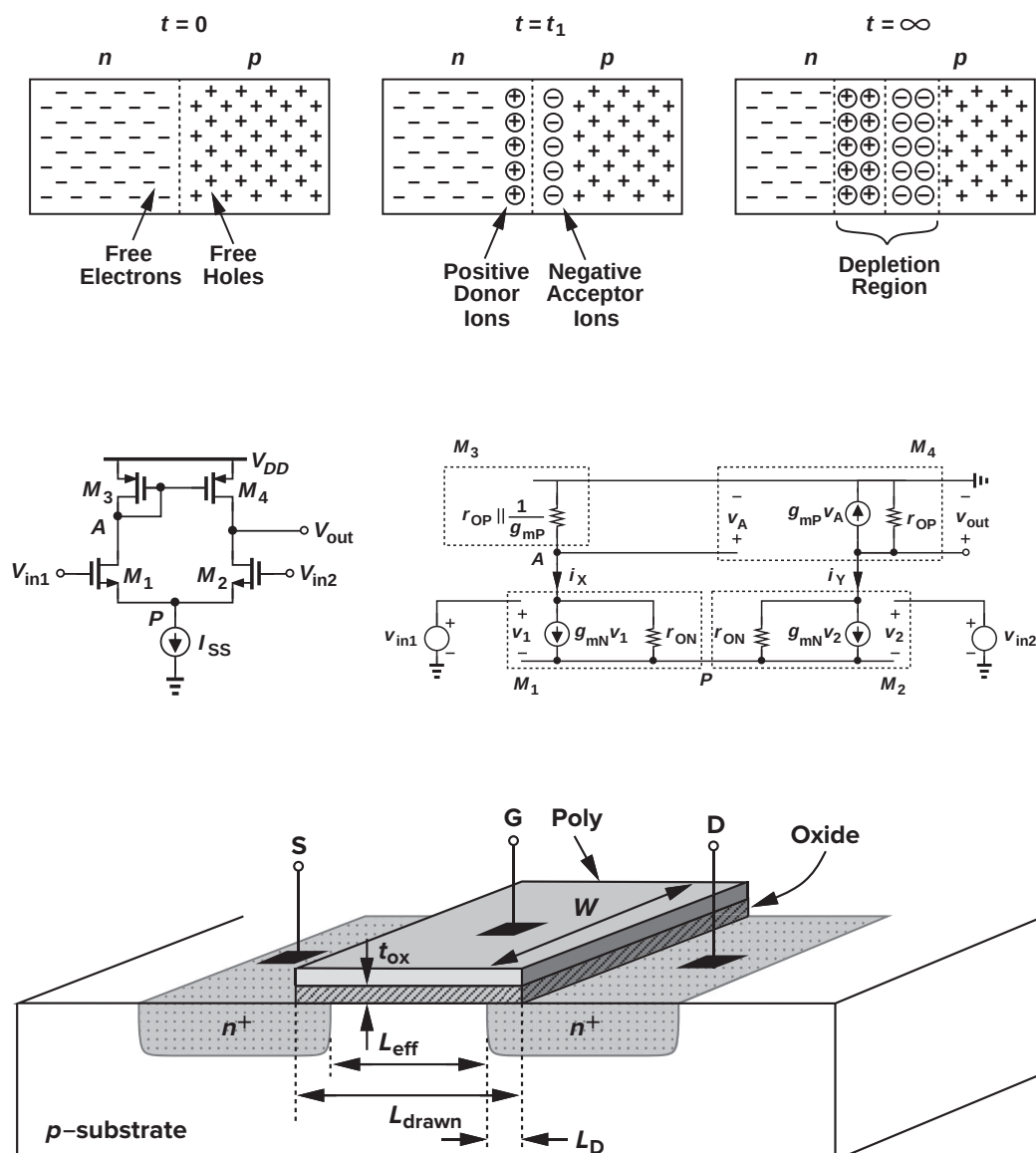
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# Preface

To be completed...

## 序言

待完成……

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# Chapter 1 Introduction to Microelectronics

**Table 1.1: Frequently used physical constant**

| Symbol          | Note                       | Value                                                        | Unit              |
|-----------------|----------------------------|--------------------------------------------------------------|-------------------|
| $k$             | Boltzmann constant         | $1.38 \times 10^{-23}$                                       | J                 |
| $c$             | velocity of light          | $2.9979 \times 10^8$                                         | m/s               |
| $h$             | velocity of light          | $6.626\,070\,15 \times 10^{-34}$                             | J · s             |
| $R$             | universal gas constant     | 8.31                                                         | J / (mol · K)     |
| $\varepsilon_0$ | permittivity of vacuum     | $8.854187817 \times 10^{-12} \approx 1/36\pi \times 10^{-9}$ | F/m               |
| $e$             | electron charge            | $1.602 \times 10^{-19}$                                      | C                 |
| $m_e$           | electronic mass            | $9.109\,382\,15\,(45) \times 10^{-31}$                       | kg                |
| $m_e$           | electronic mass            | 0.510 998 910 (13)                                           | MeV/ $c^2$        |
| $m_p$           | proton mass                | $1.672\,621\,637\,(83) \times 10^{-27}$                      | kg                |
| $m_p$           | proton mass                | 938. 272 013 (23)                                            | MeV/ $c^2$        |
| $m_p$           | proton mass                | 1. 007 276 466 77 (10)                                       | u                 |
| $m_p/m_e$       | proton-electron mass ratio | 1836.152 672 47 (80)                                         | 1                 |
| $N_A$           | avogadro's constant        | $6.02 \times 10^{23}$                                        | mol <sup>-1</sup> |
| $hc$            |                            | $1.9864 \times 10^{-25}$                                     | m · J             |
| $hc$            |                            | 1242                                                         | nm · eV           |
| $\hbar c$       |                            | 197                                                          | nm · eV           |

**Table 1.2: Frequently used physical unit conversion**

| Unit | New Unit                    |
|------|-----------------------------|
| 1 eV | $1.60218 \times 10^{-19}$ J |
| 1 J  | $6.25 \times 10^{18}$ eV    |
| 1 u  | 931.5 MeV/ $c^2$            |
| 1 u  | $1.66 \times 10^{-27}$ kg   |
| 8 Å  | $1.66 \times 10^{-27}$ kg   |

# Chapter 2 Basic Physics of Semiconductors

One cannot design a high-performance analog circuits without a detailed knowledge of the analog devices and their limitations. However, we do face a dilemma. Our treatment of device physics must contain enough depth to provide adequate understanding, but must also be sufficiently brief to allow quick entry into circuits. This chapter accomplishes this task.

In this chapter, we begin with the concept of semiconductors and study the movement of charge (i.e., the flow of current) in them. Then, we deal with the “pn junction,” which also serves as diode, and formulate its behavior.

## 2.1 Semiconductor Materials

The intrinsic concentration  $n_i$  (the number of electrons per unit volume in intrinsic silicon) is given by:

$$n_i = A \cdot T^{\frac{3}{2}} \exp \frac{-E_{g0}}{2kT} \quad (\text{electrons/cm}^3) \quad (2.1)$$

where:

- (1)  $A$  is a constant of  $5.2 \times 10^{15} \text{ cm}^{-3} \text{ K}^{-\frac{3}{2}}$  (Razavi) or  $3.88 \times 10^{16} \text{ cm}^{-3} \text{ K}^{-\frac{3}{2}}$  (Feng) for silicon. For germanium (锗 Ge),  $A = 1.76 \times 10^{16} \text{ cm}^{-3} \text{ K}^{-\frac{3}{2}}$  (Feng).
- (2)  $k = 1.38 \times 10^{-23} \text{ J/K}$  is the Boltzmann constant.
- (3)  $E_{g0}$  is the energy gap at  $T = 0 \text{ K}$ . For silicon,  $E_{g0} = 1.12 \text{ eV}$  (Razavi) or  $1.21 \text{ eV}$  (Feng); for germanium,  $E_{g0} = 0.785 \text{ eV}$  (Feng). Note  $1 \text{ eV} = 1.60218 \times 10^{-19} \text{ J}$ .

yielding:

$$\text{Razavi: } n_i = \begin{cases} 1.08 \times 10^{10} \text{ cm}^{-3}, & T = 300 \text{ K} \\ 1.54 \times 10^{15} \text{ cm}^{-3}, & T = 600 \text{ K} \end{cases} \quad (2.2)$$

The free electron concentration  $n$ , hole concentration  $p$  and n-type doping density  $N_D$  satisfy the relation:

$$np = n_i^2, \quad n = N_D + p \quad (2.3)$$

The velocity of charge carrier is proportional to the electric field:

$$v = \mu E \quad (v_e = -\mu_n E, \quad v_h = \mu_p E) \quad (2.4)$$

where the mobility  $\mu$  is given by  $\mu_n = 1350 \text{ cm}^2 \cdot \text{s}^{-1} \cdot \text{V}^{-1}$  and  $\mu_p = 480 \text{ cm}^2 \cdot \text{s}^{-1} \cdot \text{V}^{-1}$  for silicon at  $T = 300 \text{ K}$ . Note that  $\mu = \frac{\mu_0}{1 + \frac{\mu_0 E}{v_{sat}}}$  is a function of  $E$ .

We can write the drift current density as:

$$J_n = n(-q_e)(-\mu_n E), \quad J_p = pq_h \mu_p E \quad (2.5)$$

$$\Rightarrow J_{\text{drift}} = (\mu_n n + \mu_p p) q_e E \quad (2.6)$$

Equivalently, rewrite it as the electrical conductivity of the material:

$$\rho = \frac{E}{J} = \frac{1}{\sigma} \Rightarrow \sigma = \frac{J}{E} = (\mu_n n + \mu_p p) q_e \quad (2.7)$$

Concentration gradient leads to diffusion current density:

$$J_n = (-q_e) D_n \left( -\frac{dn}{dx} \right) = q_e D_n \frac{dn}{dx} \quad (2.8)$$

$$J_p = (+q_e) D_p \left( -\frac{dp}{dx} \right) = -q_e D_p \frac{dp}{dx} \quad (2.9)$$

In intrinsic silicon at  $T = 300 \text{ K}$ ,  $D_n = 34 \text{ cm}^2/\text{s}$  and  $D_p = 12 \text{ cm}^2/\text{s}$ .

## 2.2 pn Junction

### 2.2.1 pn Junction in Equilibrium

It can be proved that  $D$  and  $\mu$  satisfy the Einstein relation:

$$\frac{D}{n} = \frac{kT}{q_e} \mu = V_T \quad (2.10)$$

we call  $V_T$  the thermal voltage, which is  $26 \text{ mV}$  at  $300 \text{ K}$ . The built-in potential  $V_B$  (or denoted as  $V_0$ ) of a pn junction is given by:

$$V_B = V_T \ln \frac{N_A N_D}{n_i^2} \quad (2.11)$$

The typical value of  $V_B$  is between  $0.5 \text{ V}$  and  $0.8 \text{ V}$ . Since  $V_T$  and  $n_i$  varies with temperature,  $V_B$  decreases about  $2.5 \text{ mV}$  per  $1 \text{ K}$  increase in temperature.

As the depletion region in pn junction must be charge neutral, we have:

$$Q_- = (-q_e) S x_p N_A, \quad Q_+ = (+q_e) S x_n N_D \quad (2.12)$$

$$\Rightarrow x_n N_D = x_p N_A \quad (2.13)$$

It can be proved that the width of the depletion region is:

$$l_0 = x_n + x_p = \sqrt{\frac{2\varepsilon}{q_e} \cdot \frac{N_A + N_D}{N_A N_D} \cdot V_B} = \frac{C_{j0} V_B}{q_e} \quad (2.14)$$

where  $\varepsilon = \varepsilon_0 \varepsilon_r = 11.7 \varepsilon_0$  is the permittivity of silicon.

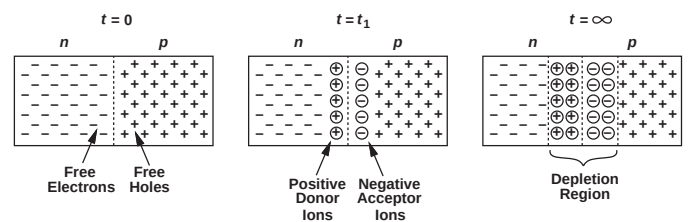


Figure 2.1: Evolution of the depletion region in a pn junction

### 2.2.2 pn Junction Under Reverse Bias

From eq (2.14), we can see the junction capacitance  $C_j$  is given by:

$$C_j = \frac{C_{j0}}{\left(1 - \frac{V_D}{V_B}\right)^n} \quad (2.15)$$

$$C_{j0} = \sqrt{\frac{\epsilon q_e}{2} \cdot \frac{N_A N_D}{N_A + N_D} \cdot \frac{1}{V_B}} \quad (2.16)$$

where  $n$  is the modified factor, which is typically  $\frac{1}{2}$  for abrupt junctions,  $\frac{1}{3}$  for linearly graded junctions and  $\frac{1}{2} \sim 6$  to super-abrupt junctions.

A low-doping junction has a relatively high breakdown voltage (avalanche breakdown), while a high-doping junction has a relatively low breakdown voltage (Zener breakdown).

### 2.2.3 pn Junction Under Forward Bias

For every  $V_T \ln 10 \approx 60$  mV increase in the forward voltage,  $I_D$  becomes ten times the original.

Note that the current-voltage characteristic is relatively sensitive to temperature, i.e., for every  $1^\circ\text{C}$  increase in temperature, the curve shifts to the left by  $2 \text{ mV} \sim 2.5 \text{ mV}$ .

In some cases, the static characteristic of a diode needs to be modified for better accuracy:

$$I_D = I_S \left[ \exp\left(\frac{V}{nV_T}\right) - 1 \right] \quad (2.17)$$

where  $r_S$  is the parasitic resistance and  $n$  (typically  $1 \sim 2$ ) is the correction factor.

### 2.2.4 Temperature Dependence

**Table 2.1: Temperature Dependence of Junction Parameters**

| Parameter          | temp coefficient |
|--------------------|------------------|
| $V_B$              | -                |
| $I_S$              | +                |
| $V_{BR,Zener}$     | -                |
| $V_{BR,Avalanche}$ | +                |



## Chapter 3 Diode Models and Circuits



# Chapter 4 Physics of Bipolar Transistors

In this chapter, we analyze the structure and operation of bipolar transistors, preparing ourselves for the study of circuits employing such devices.

we aim to understand the physics of the transistor, derive equations that represent its I/V characteristics, and develop an equivalent model that can be used in circuit analysis and design.

## 4.1 Structure of Bipolar Transistors

Note that all of the operation principles and equations described for npn transistors apply to pnp devices as well.

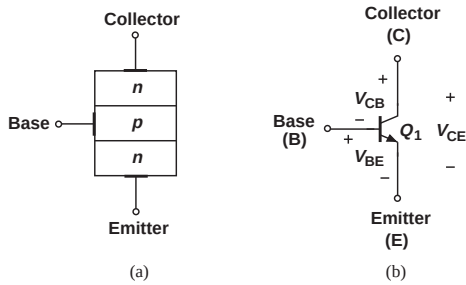


Figure 4.1: (a) Structure and (b) circuit symbol of a bipolar junction transistor (BJT)

In reality, the dimensions and doping levels of the emitter and the collector are quite different to achieve the desired characteristics. For transistors in active mode, i.e.,  $V_C > V_B > V_E$ , we have:

$$I_C = I_S \left( \exp \frac{V_{BE}}{V_T} - 1 \right) \left( 1 + \frac{V_{CE}}{V_A} \right) \quad (4.1)$$

$$\approx I_S \exp \frac{V_{BE}}{V_T} \quad (V_{BE} > 5V_T = 130 \text{ mV}) \quad (4.2)$$

where:

- (1)  $I_S = \frac{A_E q_e D_n n_i^2}{N_B W_B}$ ;
- (2)  $A_E$ : emitter cross-sectional area;
- (3)  $q_e$ : electron charge;
- (4)  $D_n$ : electron diffusion coefficient (34 cm<sup>2</sup>/s for  $D_n$ , 12 cm<sup>2</sup>/s for  $D_p$ );
- (5)  $N_B$ : base doping level;
- (6)  $W_B$ : base width;

Due to the exponential dependence of  $I_C$  on  $V_{BE}$ , for every 60 mV ( $V_T \ln 10$  at  $T = 300$  K) increase in  $V_{BE}$ ,  $I_C$  increases by a factor of 10.

In practice, the BE voltage of Integrated devices  $V_{BE,int}$  is 100 mV  $\sim$  150 mV greater than that of discrete devices  $V_{BE,dis}$ , due to the different base width and other parameters (otherwise it should be about 380mV).

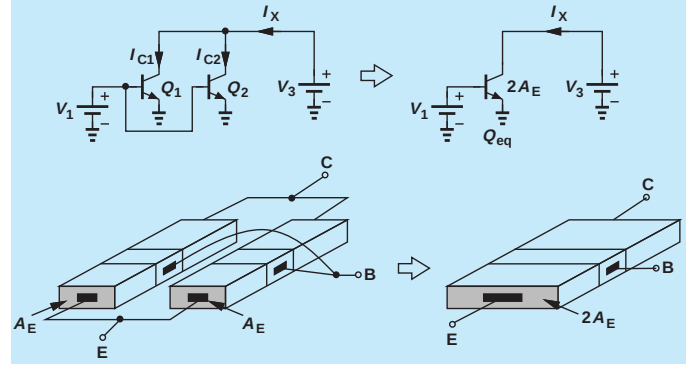


Figure 4.2: Two parallel transistors and the equivalence to a single transistor having twice area

## 4.2 Bipolar Transistor Models

For a bipolar transistor in active mode, we have the large-signal model:

$$I_C = I_S \exp \frac{V_{BE}}{V_T} \left( 1 + \frac{V_{CE}}{V_A} \right) \approx I_S \exp \frac{V_{BE}}{V_T} \quad (4.3)$$

$$I_B = \frac{I_C}{\beta}, \quad I_E = I_C + I_B = \frac{\beta + 1}{\beta} I_C = \alpha I_C \quad (4.4)$$

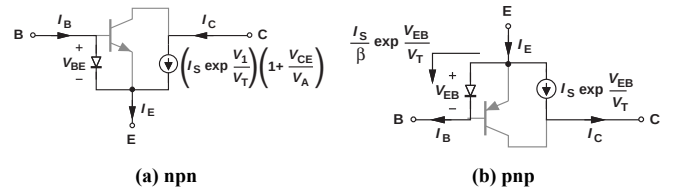


Figure 4.3: Large-signal

And the small-signal model is:

$$g_m = \frac{\partial I_C}{\partial V_{BE}} = \frac{I_C}{V_T}, \quad r_\pi = \frac{\partial V_{BE}}{\partial I_B} = \frac{\beta}{g_m} \quad (4.5)$$

$$r_O = \frac{\partial V_{CE}}{\partial I_C} = \frac{V_A + V_{CE}}{I_C} \approx \frac{V_A}{I_C} \quad (4.6)$$

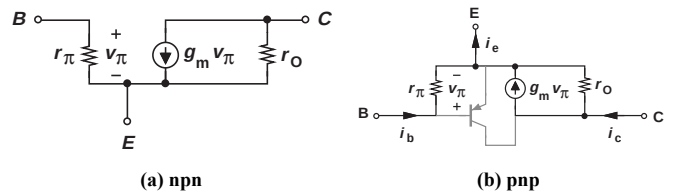
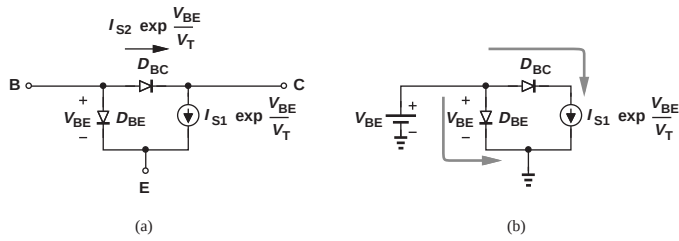


Figure 4.4: Small-signal

## 4.3 Bipolar Transistor in Saturation

The term “saturation” is used because increasing the base current in this region of operation leads to little change in the collector current.

Heavy saturation leads to a sharp rise in the base current and hence a rapid fall in  $\beta$ . In addition to a drop in  $\beta$ , the speed of bipolar transistors also degrades in saturation.



**Figure 4.5: BJT in (a) saturation region and (b) with collector open**

As a rule of thumb, we permit soft saturation with  $V_{BC} < 400$  mV because the current in the B-C junction is negligible.

# Chapter 5 Bipolar Amplifiers

With the physics and operation of bipolar transistors described in Chapter 4, we now deal with amplifier circuits employing such devices. There is an extremely wide usage of amplification in microelectronics, motivating us to master the analysis and design of such building blocks.

## 5.1 BJT's Terminal Impedances

When determining the transfer of signals from one stage to the next, the I/O impedances are usually regarded as small-signal quantities, with the tacit assumption that the signal levels are indeed small.

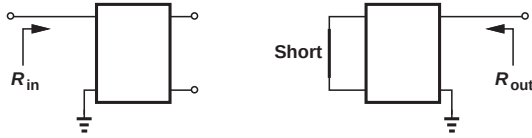


Figure 5.1: Impedance seen at a node

We summarize the small-signal impedances of the BJT in Table 5.1. Remark that the impedance here denotes to the small-signal impedance, while we use an upper case here.

## 5.2 Biasing Techniques

### 5.2.1 Simple Biasing

The biasing scheme above is simple, but suffering from the low stability and accuracy.

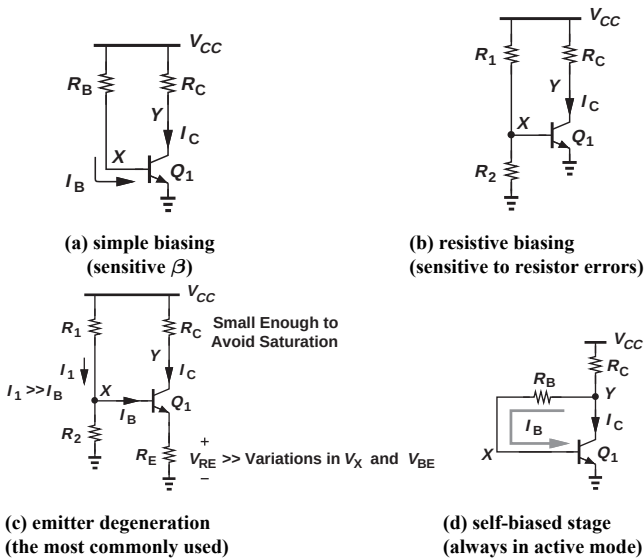


Figure 5.2: Basic biasing techniques

First, the uncertainty of  $V_{BE}$  becomes significant because the bias is sensitive to  $V_{BE}$  variations. Second,  $I_C$

heavily depends on  $\beta$ , a parameter that may change considerably with temperature and transistor mismatch. For these reasons, the topology is rarely used in practice.

$$I_B = \frac{V_{CC} - V_{BE}}{R_B}, \quad I_C = \beta I_B = \beta \frac{V_{CC} - V_{BE}}{R_B} \quad (5.1)$$

$$V_{CE} = V_{CC} - \beta \frac{V_{CC} - V_{BE}}{R_B} R_C \quad (5.2)$$

### 5.2.2 Resistive Divider Biasing

As shown in Figure 5.2 (b), the resistive divider biasing constructs a relatively stable base bias current.

$$R_{Thev} = R_1 \parallel R_2, \quad V_{Thev} = \frac{R_2 V_{CC}}{R_1 + R_2} \quad (5.3)$$

$$V_{BE} = V_{Thev} - I_B R_{Thev} \quad (5.4)$$

However, this circuit is significantly sensitive to the resistor values. A 1% error in  $R_2$  introduces a 31% ~ 36% error (typ.) in the collector current (and base current), making it hard to define the operation point. The circuit is therefore still of little practical value.

### 5.2.3 Biasing with Emitter Degeneration

In the circuit shown in Figure 5.2 (c), a relatively large variation in  $V_X$  causes negligible change in  $V_{BE}$ , hence the collector current becomes more stable.

For example, a 1% error in  $R_2$  introduces only 4% ~ 8% error (typ.) in the collector current, which is a significant improvement over the preceding circuits.

$$I_C \approx I_E = \frac{V_P}{R_E} = \frac{V_X - V_{BE}}{R_E} \quad (5.5)$$

$$\Delta I_C = \frac{\Delta V_X}{R_E} - \frac{\Delta V_{BE}}{R_E} \quad (5.6)$$

In this circuit, two rules are typically followed:

- (1)  $I_1 \gg I_B$  to lower the sensitivity to  $\beta$ , or use the Thevenin equivalent to gain a more precise bias current.
- (2)  $V_{RE}$  must be large enough (100 mV to several hundred mV) to suppress the effect of uncertainties in  $V_X$  and  $V_{BE}$  due to resistance mismatch.

### 5.2.4 Self-Biasing

Another frequently used biasing technique is the self-biasing circuit, as shown in Figure 5.2 (d).

The most important property is that it guarantees  $Q_1$  to operate in active mode (because  $V_C \geq V_B$ ), regardless of the device and circuit parameters.

$$I_C = \frac{V_{CC} - V_{BE}}{R_C + \frac{R_B}{\beta}}, \quad I_B = \frac{I_C}{\beta} \quad (5.7)$$

In this stage,  $V_{CC} - V_{BE}$  must be much greater than  $\Delta V_{BE}$  to stabilize the bias point, and  $R_C$  must be much greater than  $\frac{R_B}{\beta}$  ( $R_C > 10 \frac{R_B}{\beta}$ ) to lower the sensitivity to  $\beta$ .

Table 5.1: BJT's small-signal terminal resistances

| Collector Resistance                                                                                                                   | Base Resistance                                                                  | Emitter Resistance                                                                                                                    |
|----------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------|
| $R_{coll} = r_O \cdot \left[ 1 + \left( \frac{\beta}{r_\pi + R_B} + \frac{1}{r_O} \right) \cdot (R_E \parallel (r_\pi + R_B)) \right]$ | $R_{base} = r_\pi + R_E \cdot \frac{\beta r_O + (r_O + R_C)}{R_E + (r_O + R_C)}$ | $R_{emit} = \left( 1 + \frac{R_C}{r_O} \right) \cdot \left( \frac{r_\pi + R_B}{\beta + 1 + \frac{R_C}{r_O}} \parallel r_O \right)$    |
| $R_{coll} _{R_B=0} = r_O \cdot \left[ 1 + \left( \frac{\beta}{r_\pi} + \frac{1}{r_O} \right) \cdot (R_E \parallel r_\pi) \right]$      | $R_{base} _{R_C=0} = r_\pi + (\beta + 1)(R_E \parallel r_O)$                     | $R_{emit} _{R_C=0} = \frac{r_\pi + R_B}{\beta + 1} \parallel r_O$                                                                     |
| $R_{coll} _{R_E=0} = r_O$                                                                                                              | $R_{base} _{R_E=0} = r_\pi$                                                      | $R_{emit} _{R_B=0} = \left( 1 + \frac{R_C}{r_O} \right) \cdot \left( \frac{r_\pi}{\beta + 1 + \frac{R_C}{r_O}} \parallel r_O \right)$ |
| $R_{coll} _{r_O \rightarrow \infty} = \infty$                                                                                          | $R_{base} _{r_O \rightarrow \infty} = r_\pi + (\beta + 1)R_E$                    | $R_{emit} _{r_O \rightarrow \infty} = \frac{r_\pi + R_B}{\beta + 1}$                                                                  |

Table 5.2: Three basic types of BJT amplifiers

| Parameter                           | CE (Common Emitter)                                                                                                                                                         | CC (Common Collector, EF)                                                                            | CB (Common Base)                                                                                                        |
|-------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| $R_{out}$                           | $R_C \parallel R_{coll}$                                                                                                                                                    | $R_E \parallel R_{emit}$                                                                             | $R_C \parallel R_{coll}$                                                                                                |
| $G_m$                               | $\frac{\beta}{r_\pi + R_B} \cdot \frac{R_E \parallel \frac{r_\pi + R_B}{\beta + 1} \parallel r_O}{R_E} \approx \frac{1}{R_E + \frac{r_\pi + R_B}{\beta + 1} \parallel r_O}$ | $\frac{-g_m}{\left( 1 + \frac{R_B}{r_\pi} \right) \left( 1 + \frac{R_C}{r_O} \right)}$               | $\frac{-1}{R_E + \frac{r_\pi + R_B}{\beta + 1} \parallel r_O}$                                                          |
| $G_m $                              | $G_m _{R_E=0} = \frac{\beta}{r_\pi + R_B}$                                                                                                                                  | $G_m _{R_C=0} = \frac{-\beta}{r_\pi + R_B}$                                                          | $G_m _{R_E=0} = - \left( \frac{\beta + 1}{r_\pi + R_B} + \frac{1}{r_O} \right)$                                         |
| $G_m $                              | $G_m _{R_B=0} = g_m \cdot \frac{R_E \parallel \frac{r_\pi}{\beta + 1} \parallel r_O}{R_E}$                                                                                  | $G_m _{R_B=0} = \frac{-g_m}{1 + \frac{R_C}{r_O}}$                                                    | $G_m _{R_B=0} = \frac{-1}{R_E + \frac{r_\pi}{\beta + 1}} \parallel r_O$                                                 |
| $R_{out} _{r_O \rightarrow \infty}$ | $R_C$                                                                                                                                                                       | $R_E \parallel \frac{r_\pi + R_B}{\beta + 1}$                                                        | $R_C$                                                                                                                   |
| $G_m _{r_O \rightarrow \infty}$     | $\frac{\beta}{(\beta + 1)R_E + r_\pi + R_B} \approx \frac{1}{R_E + \frac{1}{g_m} + \frac{R_B}{\beta + 1}}$                                                                  | $\frac{-\beta}{r_\pi + R_B}$                                                                         | $\frac{-(\beta + 1)}{(\beta + 1)R_E + r_\pi + R_B} \approx \frac{-1}{R_E + \frac{1}{g_m} + \frac{R_B}{\beta + 1}}$      |
| $A_v _{r_O \rightarrow \infty}$     | $\frac{-\beta R_C}{(\beta + 1)R_E + r_\pi + R_B} \approx \frac{-R_C}{R_E + \frac{1}{g_m} + \frac{R_B}{\beta + 1}}$                                                          | $\frac{\beta R_E}{(\beta + 1)R_E + r_\pi + R_B} \approx \frac{R_E}{R_E + \frac{r_\pi + R_B}{\beta}}$ | $\frac{-(\beta + 1)R_C}{(\beta + 1)R_E + r_\pi + R_B} \approx \frac{-R_C}{R_E + \frac{1}{g_m} + \frac{R_B}{\beta + 1}}$ |

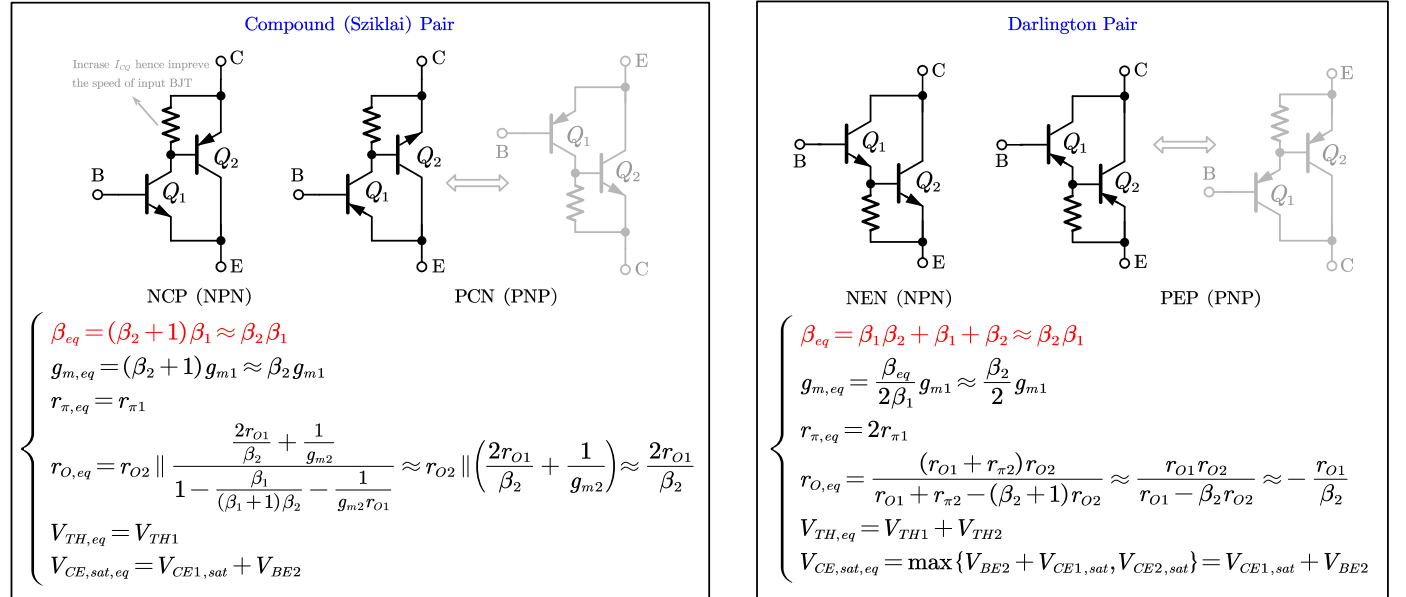
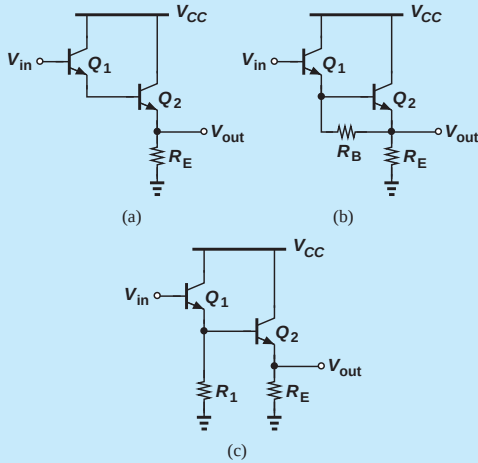


Figure 5.3: The compound (Sziklai) pair and Darlington pair

### Did you know?

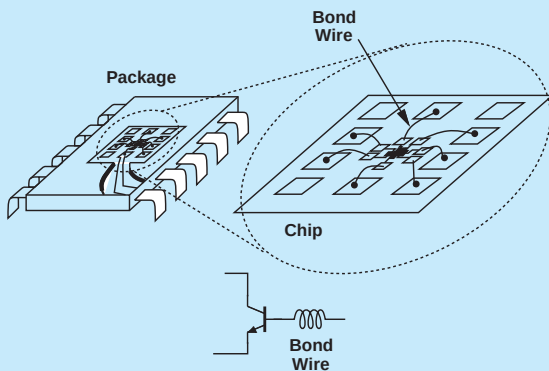
Shortly after the invention of the bipolar transistor, a Bell Labs engineer named Sidney Darlington borrowed a few transistors from his boss and took them home for the weekend. It was over that weekend that he came up with what is known as the “Darlington Pair” [Fig. (a)]. Here,  $Q_1$  and  $Q_2$  can be viewed as two emitter followers in a cascade. The circuit provides a high input impedance but since  $I_{C1}$  is small ( $= I_{B2}$ ),  $Q_1$  is quite slow. For this reason, the modified topologies in Figs. (b) and (c) are preferred.



Darlington pairs.

### Did you know?

The CB gain reduction as a result of an impedance in series with the base has stumbled many a designer. This effect is especially pronounced at high frequencies because the parasitic inductances of the package (which contains the chip) may introduce a significant impedance ( $= \omega L$ ). As shown below, the “bond wires” connecting the chip to the package behave like inductors, thus degrading the performance. For example, a 5-mm bond wire has an inductance of about 5 nH, exhibiting an impedance of 157  $\Omega$  at 5 GHz.



Effect of bond wire inductance in a CB stage.

### 5.2.5 Biasing Design Procedure

It is possible to prescribe a design procedure for the resistive divider biasing with emitter degeneration, which serves most applications.

- (1) Calculate  $I_C$  by the required small-signal quantities such as  $g_m$  and  $r_\pi$ ;
- (2) Get  $gV_{BE} = V_T \ln \frac{I_C}{I_S}$  and  $I_B = \frac{I_C}{\beta}$ , or from the datasheet;
- (3) Choose  $V_{RE} \gg \Delta V_X$ , hence  $R_E = \frac{V_{RE}}{I_C}$ ;
- (4) Calculate  $V_X = V_{BE} + V_{RE}$ ,  $R_1$  and  $R_2$ , ensuring that  $I_1 \gg I_B$  (or use the Thevenin equivalent to gain a more precise result);
- (5) Determine  $R_C$  as large as possible to obtain a high gain, but not too large to avoid saturation.

## 5.3 Bipolar Amplifier Topologies

The common parameters of the three basic types of BJT amplifiers are summarized in Table 5.2.

From the CE and CB stages, we know that  $|A_v| \approx g_m R_C$ , limited by the supply voltage (and the minimum  $V_{CE,min} = V_{BE}$ ):

$$|A_v| \approx \frac{I_C R_C}{V_T} = \frac{V_{RC}}{V_T} < \frac{V_{CC} - V_{BE}}{V_T} \quad (5.8)$$

For example, if  $V_{CC} - V_{BE} = 1$  V, then  $|A_v| < 38.5 \approx 40$ .

The intrinsic gain of the BJT is defined as:

$$A_{int} = g_m r_O = \frac{\frac{I_C}{V_T}}{\frac{V_A + V_{CE}}{I_C}} \approx \frac{V_A}{V_T} \quad (5.9)$$

In modern integrated BJT,  $V_A$  falls in the vicinity of 5 V ( $A_{int} = 200$ ), and  $A_{int}$  falls to about 50 because of a series of second-order effects.

## 5.4 Bipolar Amplifier with Biasing and Coupling Circuit

When adding the input to the amplifier, we usually use a coupling capacitor, isolating the bias condition and injecting the ac signal. In this case, we say the stage is “ac-coupled” or “capacitively coupled” (but op amp is usually dc-coupled).

The value of capacitor is chosen so that it provides a relatively low impedance (almost a short circuit) for the frequencies of interest.

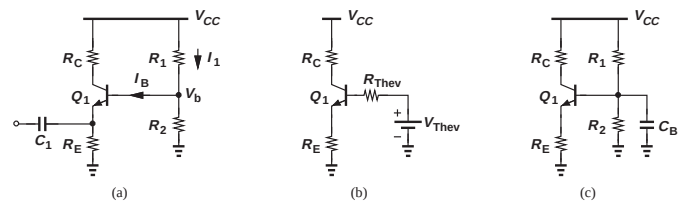


Figure 5.4: (a) CB stage with base bias network, (b) use of Thevenin equivalent, (c) effect of bypass capacitor.

# Chapter 6 Physics of MOS Transistors

See **Razavi CMOS** for more details.

## 6.1.3 Misc

$$\text{Discrete MOSFET: (three cap model)} \quad \begin{cases} C_{iss} = C_{GD} + C_{GS} \\ C_{oss} = C_{GD} + C_{DS} \\ C_{rss} = C_{GD} \end{cases} \quad (6.1)$$

## 6.1 Chapter Summary

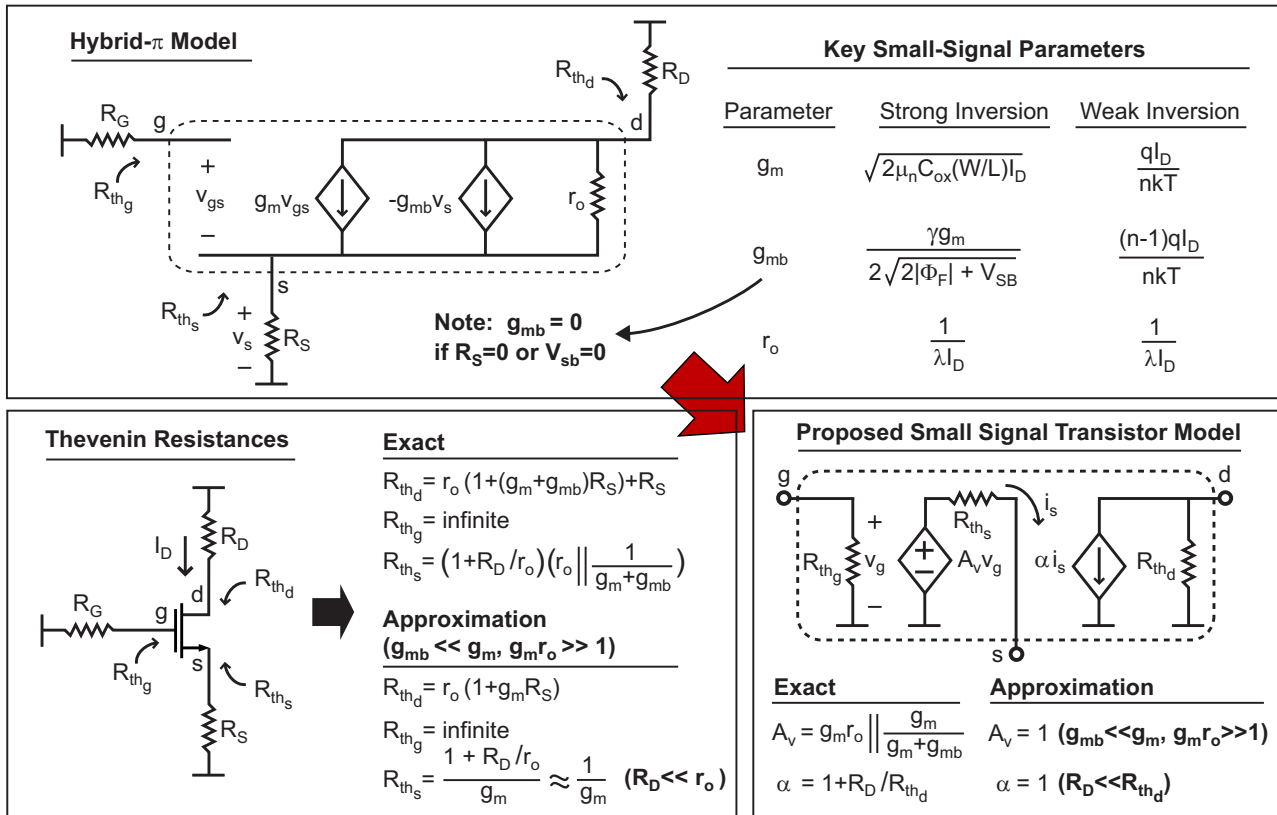
### 6.1.1 Chapter Summary of NMOS

$$\begin{aligned} V_{TH} &= V_{TH0} + \gamma \left( \sqrt{|2\Phi_F| + V_{SB}} - \sqrt{|2\Phi_F|} \right) \\ g_{mb} &= \eta g_m, \quad \eta = \frac{\partial V_{TH}}{\partial V_{SB}} = \frac{\gamma}{2\sqrt{|2\Phi_F| + V_{SB}}} \\ I_D &= \begin{cases} I_S \left[ \exp\left(\frac{V_{GS}}{nV_T}\right) - 1 \right], & \text{weak inversion} \\ \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{GS} - V_{TH})^2 (1 + \lambda V_{DS}) & \\ \mu_n C_{ox} \frac{W}{L} \left[ (V_{GS} - V_{TH}) V_{DS} - \frac{1}{2} V_{DS}^2 \right] & \end{cases} \\ I_S &= j_S (A_D + A_S), \quad \text{area of Drain and Source} \\ g_m &= \begin{cases} \sqrt{2\mu_n C_{ox} \frac{W}{L} I_D}, & \text{strong inversion} \\ \frac{I_D}{nV_T}, & \text{weak inversion} \end{cases} \\ g_{mb} &= \begin{cases} \frac{\gamma}{2\sqrt{|2\Phi_F| + V_{SB}}} \cdot g_m, & \text{strong inversion} \\ (n-1) g_m, & \text{weak inversion} \end{cases} \\ r_O &= \frac{1 + \lambda V_{DS}}{\lambda I_D} = \frac{V_A + V_{DS}}{I_D} \\ C_{GD} &= \begin{cases} WC_{ov,GD}, & \text{off, satu} \\ WC_{ov,GD} + \frac{1}{2} WLC_{ox}, & \text{triode} \end{cases} \\ C_{GS} &= \begin{cases} WC_{ov,GD}, & \text{off} \\ WC_{ov,GD} + \frac{2}{3} WLC_{ox}, & \text{satu} \\ WC_{ov,GD} + \frac{1}{2} WLC_{ox}, & \text{triode} \end{cases} \\ f_T &\approx \frac{g_m}{2\pi C_{GS} + C_{GD}} \end{aligned}$$

### 6.1.2 Chapter Summary of PMOS

$$\begin{aligned} V_{TH} &= V_{TH0} + \gamma \left( \sqrt{|2\Phi_F| - V_{SB}} - \sqrt{|2\Phi_F|} \right) \\ g_{mb} &= \eta g_m, \quad \eta = \frac{\partial V_{TH}}{\partial V_{SB}} = \frac{\gamma}{2\sqrt{|2\Phi_F| - V_{SB}}} \\ (I_D) &= \begin{cases} I_S \left[ \exp\left(\frac{V_{SG}}{nV_T}\right) - 1 \right], & \text{weak inversion} \\ \frac{1}{2}\mu_n C_{ox} \frac{W}{L} (V_{SG} - V_{TH})^2 (1 + \lambda V_{SD}) & \\ \mu_n C_{ox} \frac{W}{L} \left[ (V_{SG} - V_{TH}) V_{SD} - \frac{1}{2} V_{SD}^2 \right] & \end{cases} \\ I_S &= j_S (A_D + A_S), \quad \text{area of Drain and Source} \\ g_m &= \begin{cases} \sqrt{2\mu_n C_{ox} \frac{W}{L} (-I_D)}, & \text{strong inversion} \\ \frac{(-I_D)}{nV_T}, & \text{weak inversion} \end{cases} \\ g_{mb} &= \begin{cases} \frac{\gamma}{2\sqrt{|2\Phi_F| - V_{SB}}} \cdot g_m, & \text{strong inversion} \\ (n-1) g_m, & \text{weak inversion} \end{cases} \\ r_O &= \frac{1 + \lambda V_{SD}}{\lambda (-I_D)} = \frac{V_A + V_{SD}}{(-I_D)} \end{aligned}$$

## Summary Sheet for CMOS Thevenin Modeling





## Small Signal Models of the MOSFET Transistor

6.301 — Spring 2002

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### General Thevenin Resistance View with D-G Feedback

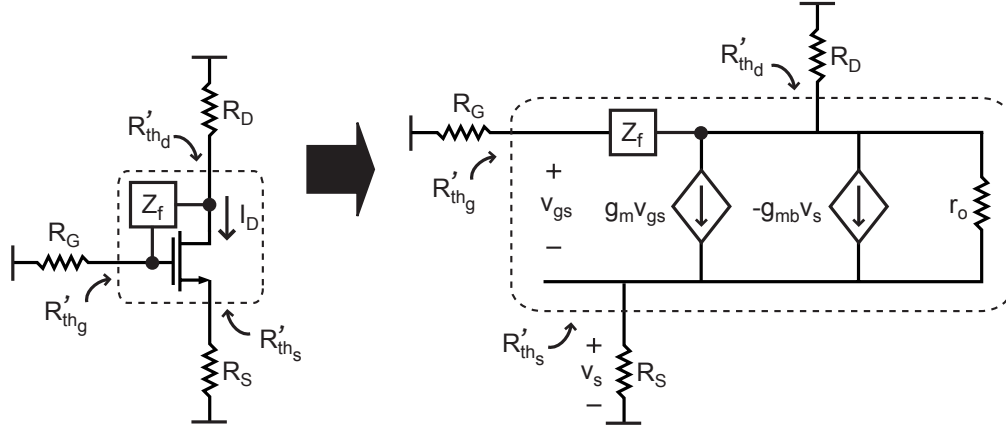


Figure 1 Definition of Thevenin Resistances.

Thevenin resistance at drain:

$$Z'_{th_d} \approx R_{th_d} \parallel \frac{Z_f}{1 + (Z_f \parallel R_G)(G_m - 1/Z_f)}$$

Thevenin resistance at gate:

$$Z'_{th_g} \approx R_{th_g} \parallel \frac{Z_f}{1 + (Z_f \parallel R_{th_d} \parallel R_D)(G_m - 1/Z_f)}$$

Thevenin resistance at source:

$$Z'_{th_s} \approx \frac{1}{g_m} + \frac{1}{1 + \beta_o} (R_B \parallel \frac{Z_f}{1 + (Z_f \parallel R_{th_c} \parallel R_C)(G_m - 1/Z_f)})$$

Where:

$$R_{th_c} \approx (1 + g_m(r_\pi \parallel R_E))r_o \quad \text{for } R_B \ll r_\pi, R_E$$

$$R_{th_b} \approx r_\pi + (\beta_o + 1)R_E \quad \text{for } R_C \ll r_o, R_E \ll r_o$$

$$G_m = \frac{1}{1/g_m + R_E}$$

Note: for small signal two-port models, simply replace the non-feedback Thevenin resistances with the  $Z'$  Thevenin impedances above, and keep the same  $G_m$  and  $A_v$  values as before.

Figure 6.2: Impedance Summary of MOS (With Feedback)  
[https://cppsim.com/CircuitLectures/impedance\\_view\\_mos\\_feedback.pdf](https://cppsim.com/CircuitLectures/impedance_view_mos_feedback.pdf)

## Small Signal Models of the Bipolar Transistor

6.301 — Spring 2002

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### General Thevenin Resistance View with C-B Feedback

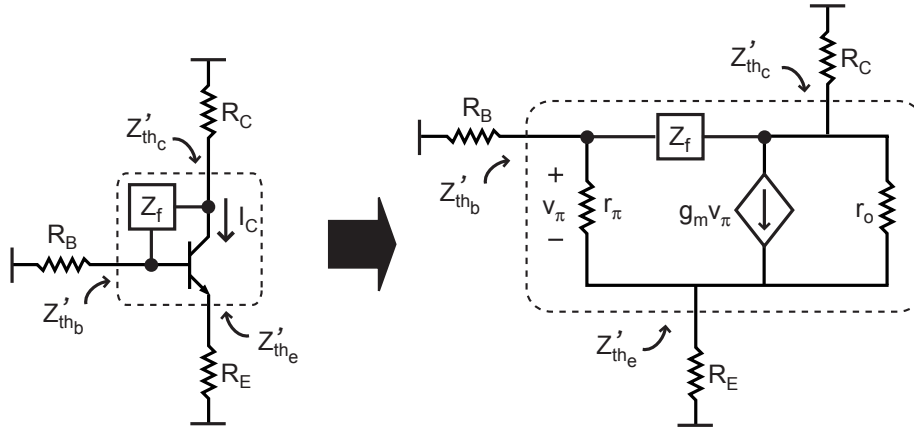


Figure 1 Definition of Thevenin Resistances.

Thevenin resistance at collector:

$$Z'_{thc} \approx R_{thc} \parallel \frac{Z_f}{(Z_f \parallel R_{thb} \parallel R_b)(G_m + 1/(R_{thb} \parallel R_b))}$$

Thevenin resistance at base:

$$Z'_{thb} \approx R_{thb} \parallel \frac{Z_f}{(Z_f \parallel R_{thc} \parallel R_c)(G_m + 1/(R_{thc} \parallel R_c))}$$

Thevenin resistance at emitter:

$$Z'_{the} \approx \frac{1}{g_m} + \frac{1}{1 + \beta_o} (R_b \parallel (R_c + Z_f)) + \frac{1}{1 + R_f/R_c} (R_b \parallel (R_c + Z_f))$$

Where:

$$R_{thc} \approx (1 + g_m(r_\pi \parallel R_E))r_o \quad \text{for } R_B \ll r_\pi, R_E$$

$$R_{thb} \approx r_\pi + (\beta_o + 1)R_E \quad \text{for } R_C \ll r_o, R_E \ll r_o$$

$$G_m = \frac{1}{1/g_m + R_E}$$

Figure 6.3: Impedance Summary of MOS (With Feedback)  
[https://cppsim.com/CircuitLectures/impedance\\_view\\_feedback.pdf](https://cppsim.com/CircuitLectures/impedance_view_feedback.pdf)

## Chapter 7 CMOS Amplifiers

See **Razavi CMOS** for more details.

**Table 7.1: Three basic types of CMOS amplifiers**

| Parameter                                                                  | CS (Common Source)                           | CD (SF, Source Follower)                                 | CG (Common Gate)                                            |
|----------------------------------------------------------------------------|----------------------------------------------|----------------------------------------------------------|-------------------------------------------------------------|
| $R_{out}$                                                                  | $R_D \parallel R_{drain}$                    | $R_S \parallel R_{source}$                               | $R_D \parallel R_{drain}$                                   |
| $G_m$                                                                      | $\frac{g_m}{1 + \frac{R_S}{R_{S0}}}$         | $\frac{-g_m}{1 + \frac{R_D}{r_O}}$                       | $\frac{-1}{R_S + R_{S0}}$                                   |
| $A_v$                                                                      | $-g_m r_O \cdot \frac{R_D}{R_D + R_{drain}}$ | $g_m R_{S0} \cdot \frac{R_S}{R_S + R_{source}}$          | $\frac{R_D \parallel R_{drain}}{R_S + R_{source}}$          |
| $\lim_{r_O \rightarrow \infty} R_{out}$                                    | $R_D$                                        | $R_S \parallel \frac{1}{g_m} \parallel \frac{1}{g_{mb}}$ | $R_D$                                                       |
| $\lim_{r_O \rightarrow \infty} G_m$                                        | $\frac{1}{(1+\eta)R_S + \frac{1}{g_m}}$      | $-g_m$                                                   | $\frac{-1}{R_S + \frac{1}{g_m + g_{mb}}}$                   |
| $\lim_{r_O \rightarrow \infty} A_v$                                        | $\frac{-R_D}{(1+\eta)R_S + \frac{1}{g_m}}$   | $\frac{R_S}{(1+\eta)R_S + \frac{1}{g_m}}$                | $\frac{R_D}{R_S + \frac{1}{(1+\eta)g_m}}$                   |
| $\lim_{\substack{g_{mb} \rightarrow 0 \\ r_O \rightarrow \infty}} R_{out}$ | $R_D$                                        | $R_S \parallel \frac{1}{g_m}$                            | $R_D$                                                       |
| $\lim_{\substack{g_{mb} \rightarrow 0 \\ r_O \rightarrow \infty}} G_m$     | $\frac{g_m}{1 + g_m R_S}$                    | $-g_m$                                                   | $\frac{-1}{R_S + \frac{1}{g_m}} = \frac{-g_m}{1 + g_m R_S}$ |
| $\lim_{\substack{g_{mb} \rightarrow 0 \\ r_O \rightarrow \infty}} A_v$     | $\frac{-R_D}{R_S + \frac{1}{g_m}}$           | $\frac{R_S}{R_S + \frac{1}{g_m}}$                        | $\frac{R_D}{R_S + \frac{1}{g_m}}$                           |

## Chapter 8    Operation Amplifier as a Black Box

To be completed.

# Chapter 9 Cascode Stages and Current Mirrors

The "cascode" stage is a modified version of common-emitter (or common-source) topologies and proves useful in high-performance circuit design, and the "current mirror" is an interesting and versatile technique employed extensively in integrated circuits.

## 9.1 Cascode Stage

### 9.1.1 Cascode as a Source (Cascode Load)

We have already noticed that emitter (or source) degeneration boosts the small-signal impedance seen at the collector (or drain). For the circuits shown in Figure 9.1, we have:

$$\begin{aligned} R_{out1} &= r_O + (1 + g_m r_O)(R_E \parallel r_\pi) \approx g_m r_O (R_E \parallel r_\pi) \\ R_{out2} &= r_O + (1 + g_m r_O) R_S \approx g_m r_O R_S \end{aligned}$$

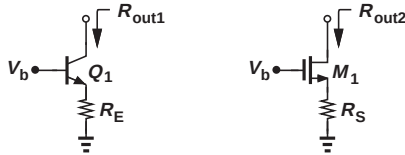


Figure 9.1: Output impedance of degenerated transistors

The cascode<sup>①</sup> stage, i.e., replacing the degeneration resistance with a biased transistor in Figure 9.1, exhibits a small-signal impedance of:

$$R_{out,BJT} \approx (g_{m1} r_{O1})(r_{O2} \parallel r_{\pi1}) \quad (9.1)$$

$$R_{out,MOS} \approx (g_{m1} r_{O1}) r_{O2} \quad (9.2)$$

It is interesting to note that the maximum  $R_{out}$  provided by a bipolar cascode is  $R_{out,max} \approx g_{m1} r_{O1} r_{\pi1} = \beta_1 r_{O1}$ .

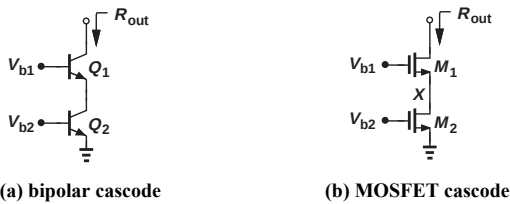


Figure 9.2: Cascode current source

### 9.1.2 Cascode as an Amplifier (Cascode Input)

The high voltage gain of the cascode topology shown in Figure 9.3 makes it attractive for many applications. In Figure 9.3 (b), the overall voltage gain is given by:

$$R_{out} = R_{out,p} \parallel R_{out,n} \quad (9.3)$$

$$R_{out,p} \approx g_{m3} r_{O3}(r_{\pi3} \parallel r_{O4}) \quad (9.4)$$

$$R_{out,n} \approx g_{m2} r_{O2}(r_{\pi2} \parallel r_{O1}) \quad (9.5)$$

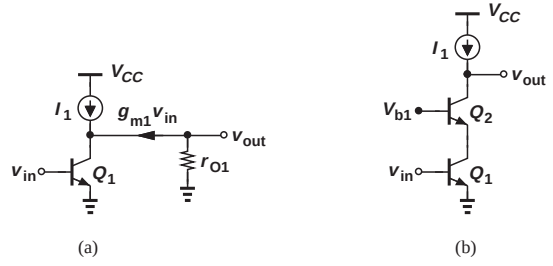


Figure 9.3: Cascode stage as an amplifier: (a) flow of output current generated by a CE stage through  $r_{O1}$ , (b) use of cascode to increase the output impedance.

Acting as a pure resistance,  $Q_2$  does not affect the transconductance since its base and collector are tied to ac-ground (in  $G_m$  calculation). Therefore, we still have:

$$G_m = \frac{r_{O1} \parallel r_{\pi2}}{r_{O1} \parallel r_{\pi2} + \frac{1}{g_{m2}} \parallel r_{O2}} \cdot g_{m1} \quad (9.6)$$

$$A_v = -G_m R_{out} \approx -g_{m1}(r_{\pi2} \parallel r_{O1}) \cdot (g_{m1} r_{O1}) \quad (9.7)$$

However, the biasing circuit of cascode amplifier can be kind of complex, limiting its applications in reality.

### 9.1.3 Cascode Stage

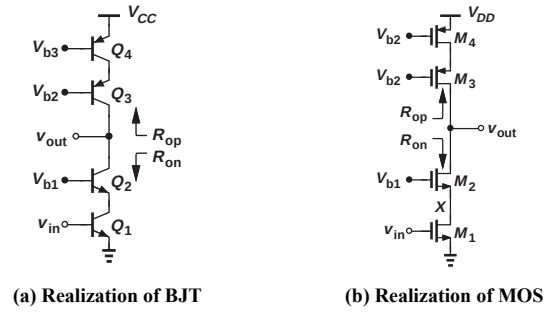


Figure 9.4: Cascode Stage

$$A_v \approx -g_{m1}(R_{out,n} \parallel R_{out,p}) \quad (9.8)$$

$$R_{out,n} \approx g_{m2} r_{O2}(r_{\pi2} \parallel r_{O1}) \quad (9.9)$$

$$R_{out,p} \approx g_{m3} r_{O3}(r_{\pi3} \parallel r_{O4}) \quad (9.10)$$

### Did you know?

The cascode stage was originally invented to suppress an undesirable high-frequency phenomenon called the "Miller effect" (Chapter 11). In fact, many discrete bipolar amplifiers still employ cascoding for the same reason. However, it was recognized in early days of CMOS technology that cascodes could also provide a high output impedance and a high voltage gain, hence their widespread use.

<sup>①</sup>The term "cascode" is believed to be an abbreviation of "cascaded triodes".

# Chapter 10 Differential Amplifiers

This chapter describes bipolar and MOS differential amplifiers and formulates their large-signal and small-signal properties.

and  $R_{SS} \gg \frac{1}{g_{mN}}$ , it can be proved that<sup>①</sup>:

$$R_{out} \approx r_{ON} \parallel r_{OP} \quad (10.4)$$

$$A_{DM} \approx g_{mN} (r_{ON} \parallel r_{OP}) \quad (10.5)$$

$$A_{CM} \approx \frac{-g_{mN} r_{ON}}{2A_{DM} \cdot g_{mP} R_{SS}} \quad (10.6)$$

$$CMRR \approx 2A_{DM}^2 \cdot \frac{g_{mP} R_{SS}}{g_{mN} r_{ON}} < 2A_{DM} (g_{mP} R_{SS}) \quad (10.7)$$

## 10.1 The Five-Transistor OTA

### 10.1.1 Large-Signal Analysis

Different from differential amplifiers, though  $V_{in,CM}$  has negligible effect on small-signal gain for the five-transistor OTA, it does affect the output swing. Concretely speaking, the input and output range is limited by:

$$V_{out} \in [V_{in,CM} + V_{TH}, V_{DD}] \quad (10.1)$$

$$V_{in,CM} < V_{OV}(I_{SS}) + V_{GS}(\frac{1}{2}I_{SS}) \quad (10.2)$$

$$V_{in,CM} > V_{DD} - V_{OV}(\frac{1}{2}I_{SS}) \quad (10.3)$$

where  $V_{OV} = V_{OV}(I_{SS})$  denotes the overdrive voltage at  $I_D = I_{SS}$ .

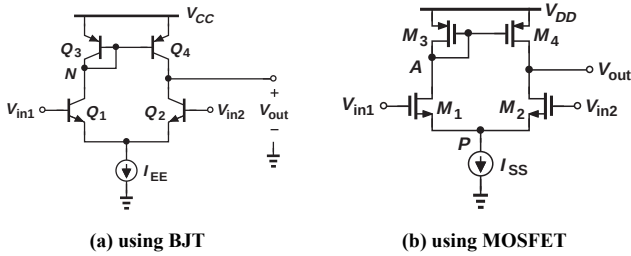


Figure 10.1: Differential pair with active load

### 10.1.2 Small-Signal Analysis

Below is the small-signal equivalent for the five-transistor OTA (ignoring  $R_{SS}$ ):

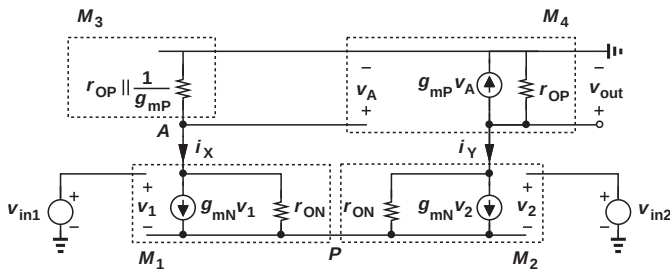


Figure 10.2: Small-signal equivalent for MOS OTA

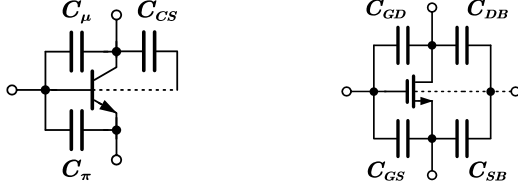
For a MOS OTA, assuming  $g_{mP} r_{OP}, g_{mN} r_{ON} \gg 1$

<sup>①</sup> see [YiDingg's Website > Blogs > The Five-Transistor OTA](https://yidingg.github.io/YiDingg/#/Blogs/Electronics/The%20Five-Transistor%20OTA) (<https://yidingg.github.io/YiDingg/#/Blogs/Electronics/The%20Five-Transistor%20OTA>)

# Chapter 11 Frequency Response

To be completed.

## 11.1 High-Frequency Models



(a) High frequency model of BJT (b) High frequency model of MOSFET

Figure 11.1: High frequency model of BJT and MOSFET

$$\text{MOS: } \begin{cases} \omega_T = \sqrt{\frac{g_m^2}{C_{GS}^2 + 2C_{GS}C_{GD}}} \approx \frac{g_m}{C_{GS} + C_{GD}} \\ f_T = \frac{\omega_T}{2\pi} \approx \frac{g_m}{2\pi C_{GS} + C_{GD}} \\ C_{GS} > \frac{2}{3}WLC_{ox} \Rightarrow f_T < \frac{3\mu_n(V_{GS} - V_{TH})}{4\pi L^2} \end{cases}$$

$$\text{BJT: } \begin{cases} \omega_T = \sqrt{\frac{g_m^2 - \frac{1}{r_{\pi}^2}}{C_{\pi}^2 + 2C_{\pi}C_{\mu}}} \approx \frac{g_m}{C_{\pi} + C_{\mu}} \\ f_T = \frac{\omega_T}{2\pi} \approx \frac{g_m}{2\pi C_{\pi} + C_{\mu}} \end{cases}$$

## 11.2 Transfer Function Analysis Techniques

### 11.2.1 Node-Pole Theorem

The node-pole theorem can be generalized as follows:

**Node-pole theorem:** if node  $j$  in the signal path exhibits a small-signal resistance of  $R_j$  to ground and a capacitance of  $C_j$  to ground, then it contributes a pole of  $\omega_p = \frac{1}{R_j C_j}$  to the transfer function.

As depicted in Figure 11.2, the overall transfer function of cascade amplifiers can be written as:

$$\frac{V_{out}}{V_{in}}(s) = \frac{A_1}{1 + R_S C_{in} \cdot s} \cdot \frac{A_2}{1 + R_1 C_N \cdot s} \cdot \frac{1}{1 + R_2 C_P \cdot s}$$

We say each node in the circuit contributes to one pole.

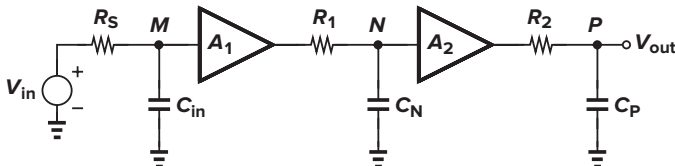


Figure 11.2: Cascade of amplifiers and the overall poles

### 11.2.2 Miller's Theorem

$$Z_1 = (1 - A_v)Z, \quad Z_2 = \left(1 - \frac{1}{A_v}\right)Z, \quad A_v = \frac{V_Y}{V_X} \quad (11.1)$$

$$\frac{1}{Y_1} = (1 - A_I)\frac{1}{Y}, \quad \frac{1}{Y_2} = \left(1 - \frac{1}{A_I}\right)\frac{1}{Y}, \quad A_I = \frac{I_2}{I_1} \quad (11.2)$$

$$Y_1 = \frac{Y}{1 - A_I}, \quad Y_2 = \frac{Y}{1 - \frac{1}{A_I}}, \quad A_I = \frac{I_2}{I_1} \quad (11.3)$$

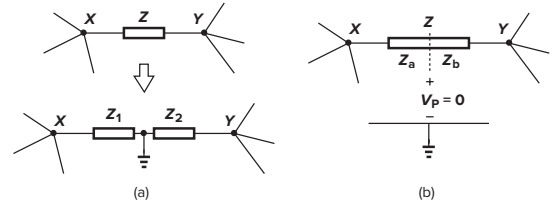


Figure 11.3: Miller's Theorem: (a) illustration of Miller's theorem, (b) identifying a local zero potential along  $Z$ .

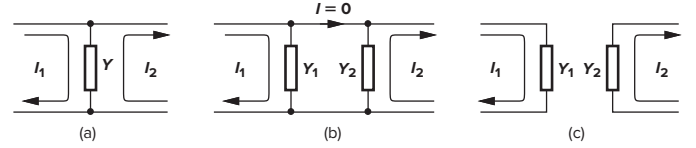


Figure 11.4: Dual of Miller's Theorem: (a) Two loops sharing admittance  $Y$ , (b) decomposition of  $Y$  into  $Y_1$  and  $Y_2$  such that  $I = 0$ , (c) equivalent circuit.

### 11.2.3 Extra Element Theorem (EET)

### 11.2.4 Tow/N Extra Element Theorem (TEET/NEET)

### 11.2.5 Zero-Value Time Constant Method (ZVTC)

### 11.2.6 Open-Circuit Time Const. Method (OCTC)

### 11.2.7 Short-Circuit Time Const. Method (SCTC)



## 11.3 Chapter Summary

$$\text{GBW} := A_0 \omega_p, \quad \frac{A_0}{\sqrt{1 + \left(\frac{\omega_u}{\omega_p}\right)^2}} = 1 \quad (11.4)$$

$$\Rightarrow \omega_u = \sqrt{A_0^2 - 1} \cdot \omega_p \approx \sqrt{A_0} \omega_p \quad (11.5)$$

$$\text{Two: } \begin{cases} \omega_{p2} = \sqrt{\sqrt{2} - 1} \cdot \omega_p = 0.64 \omega_p \\ \text{GBW}_2 = A_0^2 \omega_2 = (0.64 A_0) \cdot A_0 \omega_p \end{cases} \quad (11.6)$$

$$\text{N stages: } \begin{cases} \omega_{pn} = \sqrt{2^{\frac{1}{n}} - 1} \omega_p \\ \text{GBW}_n = A_0^n \sqrt{2^{\frac{1}{n}} - 1} \omega_p \\ = A_0^{n-1} \sqrt{2^{\frac{1}{n}} - 1} \cdot A_0 \omega_p \end{cases} \quad (11.7)$$

$$\lim_{n \rightarrow \infty} \text{GBW}_n = \begin{cases} \infty, & \text{if } A_0 > 1 \\ 0, & \text{if } A_0 \leq 1 \end{cases} \quad (11.8)$$

$$\text{High pass: } \begin{cases} \omega_{p2} = \frac{\omega_p}{\sqrt{\sqrt{2} - 1}} = 1.55 \omega_p \\ \omega_{pn} = \frac{\omega_p}{\sqrt{2^{\frac{1}{n}} - 1}} \end{cases} \quad (11.9)$$

$$\begin{cases} \omega_L \approx 1.1 \sqrt{\omega_{L1}^2 + \cdots \omega_{Ln}^2} \\ \frac{1}{\omega_H} \approx 1.1 \sqrt{\frac{1}{\omega_{H1}^2} + \cdots \frac{1}{\omega_{Hn}^2}} \end{cases} \quad (11.10)$$

## 1. CS/CE

Low Frequency Analysis

(assuming other caps open-circuit)

neglecting  $r_o$  unless otherwise specified

High Frequency Analysis

(assuming  $\frac{1}{sC_b} \parallel R_{SS} \rightarrow 0$ ) hence source shorted to ac ground and no body effect

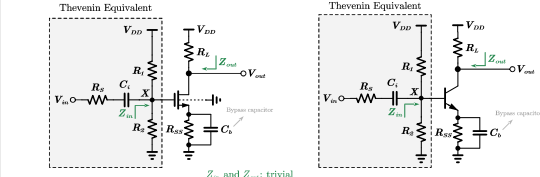
High Frequency Analysis using Miller Approximation:

$$\begin{cases} V_{out} = -g_m R_L \cdot \frac{1}{1 + s\tau_1} \cdot \frac{1}{1 + s\tau_2} \\ \tau_1 = R_{Thes} [C_{in} + (1 + g_m R_L) C_{GD}] \\ \tau_2 = R_L [C_{out} + (1 + \frac{1}{g_m R_L}) C_{DB}] \end{cases}$$

$Z_{in}$  and  $Z_{out}$ : trivial

$$\begin{cases} Z_{in} = \frac{1}{sC_{GS}} \parallel \frac{1 + sR_L(C_{GD} + C_{DB})}{sC_{DB}(1 + g_m R_L + sR_L C_{DB})} \\ Z_{out} = R_L \parallel \frac{1}{sC_{DB}} \parallel \frac{1 + sR_L(C_{GS} + C_{GD})}{sC_{DB}(1 + g_m R_L + sR_L C_{DB})} \end{cases}$$

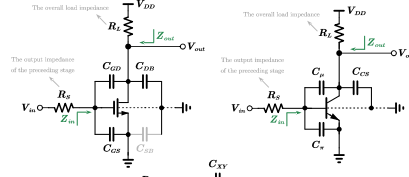
$$\begin{cases} Z_{in} = r_\pi \parallel \frac{1}{sC_{\pi}} \parallel \frac{1 + sR_L(C_{\pi} + C_{CS})}{sC_{\pi}(1 + g_m R_L + sR_L C_{CS})} \\ Z_{out} = R_L \parallel \frac{1}{sC_{CS}} \parallel \frac{1 + s(R_L \parallel r_o)(C_e + C_o)}{sC_{CS}(1 + g_m(R_L \parallel r_o) + s(R_L \parallel r_o)C_e)} \end{cases}$$



$$R_{Thes} = (R_1 + \frac{1}{sC_1}) \parallel R_2, A_{v1} = \frac{V_{Thes}}{V_m} = \frac{R_1 \parallel R_2}{R_1 \parallel R_2 + R_s + \frac{1}{sC_1}} = \frac{s(R_1 \parallel R_2)C_1}{1 + s(R_1 \parallel R_2 + R_s)C_1} \begin{cases} z=0 \\ p = -\frac{1}{(R_1 \parallel R_2 + R_s)C_1} \end{cases}$$

$$\text{MOS: } \frac{-R_L}{(1 + \eta)R_{SS} + \frac{1}{g_m}} \rightarrow A_{v2} = \frac{-R_L}{R_{SS} \parallel \frac{1}{g_m} + \frac{1}{g_m}} = \frac{-g_m R_L}{1 + (g_m + g_{mb})R_{SS}} \begin{cases} z = -\frac{1}{R_{SS}C_1} \\ p = -\frac{1 + (g_m + g_{mb})R_{SS}}{R_{SS}C_1} \end{cases}$$

$$\text{BJT: } \frac{-R_L}{R_{SS} + \frac{1}{g_m} + \frac{R_o}{\beta + 1}} \rightarrow A_{v2} = \frac{-R_L}{R_{SS} \parallel \frac{1}{g_m} + \frac{1}{g_m} + \frac{R_o}{\beta + 1}} = \frac{-g_m R_L}{1 + \frac{1 + \beta R_o}{\beta + 1} g_m R_{SS}} \begin{cases} z = -\frac{1}{R_{SS}C_1} \\ p = -\frac{1 + \beta R_o}{R_{SS}C_1} \end{cases}$$



$$\text{MOS: } V_{Thes} = V_m, R_{Thes} = R_s, R_{Thes} = R_s \parallel \frac{1}{sC_b} \parallel \frac{1}{sC_{b1}} \rightarrow A_{v1} = \frac{V_{Thes}}{V_m} = \frac{R_s}{R_s + \frac{1}{sC_b} + \frac{1}{sC_{b1}}} \begin{cases} z = -\frac{1}{C_b} \\ p_1 = -\frac{1}{R_s C_b} \\ p_2 = -\frac{1}{R_s C_{b1}} \end{cases}$$

$$\text{BJT: } V_{Thes} = V_m, R_{Thes} = R_s, R_{Thes} = R_s \parallel \frac{1}{sC_b} \parallel \frac{1}{sC_{b1}} \rightarrow A_{v1} = \frac{V_{Thes}}{V_m} = \frac{R_s}{R_s + \frac{1}{sC_b} + \frac{1}{sC_{b1}}} \begin{cases} z = -\frac{1}{C_b} \\ p_1 = -\frac{1}{R_s C_b} \\ p_2 = -\frac{1}{R_s C_{b1}} \end{cases}$$

$$\text{dominant pole approximation (holds only if } |p_1| \ll |p_2| \text{)} \begin{cases} z = -\frac{1}{C_b} \\ p_1 = -\frac{1}{R_s C_b} \\ p_2 = -\frac{1}{R_s C_{b1}} \end{cases} \rightarrow \begin{cases} s_1 = \frac{-b + \sqrt{b^2 - 4a}}{2a} \approx -\frac{1}{b} \\ s_2 = \frac{-b - \sqrt{b^2 - 4a}}{2a} \approx -\frac{b}{a} \end{cases}$$

## 2. CG/CB

Low Frequency Analysis

(assuming other caps open-circuit)

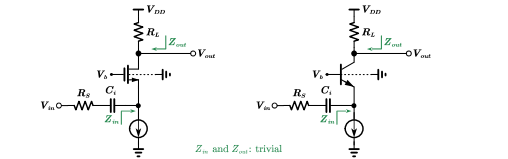
neglecting  $r_o$  unless otherwise specified

High Frequency Analysis

High Frequency Analysis Considering  $r_o$  but ignoring  $g_{mb}$

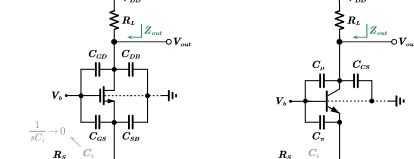
using the result from Razavi CMOS page 194 and substitute  $R_L \parallel \frac{1}{sC_L}$  for  $\frac{1}{sC_L}$

$$\begin{cases} A_v = \frac{V_{out}}{V_m} = \frac{1 + g_m r_o}{A s^2 + B s + C} \begin{cases} p_1 \approx -\frac{1}{R_s} \\ p_2 \approx -\frac{1}{R_L} \end{cases} \\ A = R_s r_o C_{out} \\ B = (1 + \frac{r_o}{R_L}) R_s C_{in} + [r_o + (1 + g_m r_o) R_L] C_{out} \\ C = 1 + \frac{r_o}{R_L} (1 + g_m r_o) R_L \end{cases}$$



$$\text{MOS: } \frac{-R_L}{R_{SS} + \frac{1}{(1 + \eta)g_m}} \rightarrow A_{v2} = \frac{-R_L}{R_{SS} + \frac{1}{(1 + \eta)g_m}} = \frac{-sR_L C_1}{1 + s(R_{SS} + \frac{1}{(1 + \eta)g_m})C_1} \begin{cases} z=0 \\ p = -\frac{1}{(R_{SS} + \frac{1}{(1 + \eta)g_m})C_1} \end{cases}$$

$$\text{BJT: } \frac{-R_L}{R_{SS} + \frac{1}{g_m} + \frac{R_o}{\beta + 1}} \rightarrow A_{v2} = \frac{-R_L}{R_{SS} + \frac{1}{g_m} + \frac{R_o}{\beta + 1}} = \frac{-sR_L C_1}{1 + s(R_{SS} + \frac{1}{g_m} + \frac{R_o}{\beta + 1})C_1} \begin{cases} z=0 \\ p = -\frac{1}{(R_{SS} + \frac{1}{g_m} + \frac{R_o}{\beta + 1})C_1} \end{cases}$$



$$\text{using node pole theorem: } A_v = \frac{V_{out}}{V_m} = A_{v0} \cdot \frac{1}{1 - \frac{1}{p_1} - \frac{1}{p_2}} \text{ at the order of } \omega_T \text{ (usually } p_{in} \gg p_{out} \text{)}$$

$$\text{MOS: } A_{v0} = \frac{-R_L}{R_{SS} + \frac{1}{(1 + \eta)g_m}} \begin{cases} C_{in} = C_{GS} + C_{GD} \\ C_{out} = C_{DB} + C_{CS} \end{cases} \begin{cases} p_{in} = -\frac{1}{(R_s \parallel R_{Thes})C_{in}} \\ p_{out} = -\frac{1}{(R_L \parallel R_{Thes})C_{out}} \end{cases}$$

$$\text{BJT: } A_{v0} = \frac{-R_L}{R_{SS} + \frac{1}{g_m} + \frac{R_o}{\beta + 1}} \begin{cases} C_{in} = C_\pi \\ C_{out} = C_e + C_{CS} \end{cases} \begin{cases} p_{in} = -\frac{1}{(R_s \parallel R_{Thes})C_{in}} \\ p_{out} = -\frac{1}{(R_L \parallel R_{Thes})C_{out}} \end{cases}$$

## 3. CD/CC

Low Frequency Analysis

(assuming other caps open-circuit)

neglecting  $r_o$  unless otherwise specified

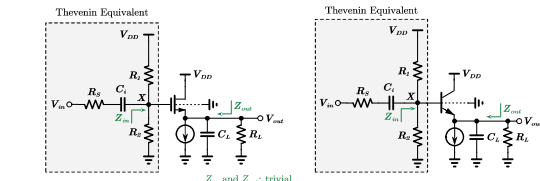
High Frequency Analysis

Input/Output Impedance Analysis:

$$A_{v, \text{midband}} \approx \frac{R_L}{R_1 + \frac{1}{g_m}} \rightarrow \begin{cases} C_x = (1 - A_v)C_{GS} = \frac{C_{GS}}{1 + g_m R_L} \\ C_y = (1 - \frac{1}{A_v})C_{CS} = \frac{C_{CS}}{g_m R_L} \end{cases}$$

$$\begin{cases} Z_{in} = \frac{1}{s(C_{GS} + C_x)} \rightarrow \text{calculations of } Z_{in}: \text{precision analysis} \\ Z_{out} = \frac{sC_{GS}(R_L \parallel \frac{1}{sC_L}) + 1}{sC_{GS} + g_m} \rightarrow \text{calculations of } Z_{out}: \text{using Miller approximation} \end{cases}$$

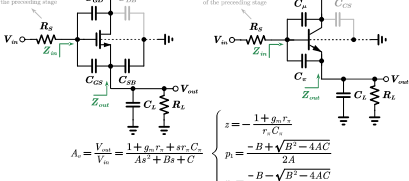
$$\begin{cases} Z_{in} = \frac{1}{s(C_\pi + C_x)} \parallel R_{Thes} = \frac{1}{s(C_\pi + C_x)} \left[ r_\pi + (\beta + 1) \left( \frac{1}{s(C_L + C_y)} \parallel R_L \right) \right] \\ Z_{out} = \frac{s r_\pi C_L (R_L \parallel \frac{1}{sC_L}) + r_\pi + R_s}{s r_\pi C_L + \beta + 1} \approx \frac{s r_\pi C_L R_L + r_\pi + R_s}{s r_\pi C_L + \beta + 1} \end{cases}$$



$$R_{Thes} = (R_1 + \frac{1}{sC_1}) \parallel R_2, A_{v1} = \frac{V_{Thes}}{V_m} = \frac{R_1 \parallel R_2}{R_1 \parallel R_2 + R_s + \frac{1}{sC_1}} = \frac{s(R_1 \parallel R_2)C_1}{1 + s(R_1 \parallel R_2 + R_s)C_1} \begin{cases} z=0 \\ p = -\frac{1}{(R_1 \parallel R_2 + R_s)C_1} \end{cases}$$

$$\text{MOS: } g_m R_{Thes} \cdot \frac{R_L}{R_s + R_{Thes}} \rightarrow A_{v2} = g_m R_{Thes} \cdot \frac{R_L}{R_s + R_{Thes}} = \frac{g_m R_{Thes}}{1 + \frac{R_{Thes}}{R_s}} \begin{cases} z = -\frac{1 + \frac{R_{Thes}}{R_s}}{R_{Thes}C_1} \\ p = -\frac{1}{R_{Thes}C_1} \end{cases}$$

$$\text{BJT: } \frac{R_s}{R_s + \frac{1}{g_m} + \frac{R_o}{\beta + 1}} \rightarrow A_{v2} = \frac{R_s}{R_s + \frac{1}{g_m} + \frac{R_o}{\beta + 1}} = \frac{R_s}{R_s + \frac{1}{g_m} + \frac{R_o}{\beta + 1}} \begin{cases} z = -\frac{1}{R_s} \\ p = -\frac{1 + \frac{1}{\beta + 1}}{C_1} \end{cases}$$



$$A_v = \frac{V_{out}}{V_m} = \frac{1 + g_m r_o}{A s^2 + B s + C} \begin{cases} z = -\frac{1 + g_m r_o}{r_o C_e} \\ p_1 = \frac{-B + \sqrt{B^2 - 4AC}}{2A} \\ p_2 = \frac{-B - \sqrt{B^2 - 4AC}}{2A} \end{cases}$$

The pole dominant approximation is usually invalid here

$$\begin{cases} \text{MOS: } \begin{cases} A = R_s (C_{GS}C_{in} + C_{GS}(C_L + C_{in}) + C_{in}(C_L + C_{in})) \\ B = g_m R_s C_{GS} + C_{GS} + C_L + C_{in} + \frac{C_{GS}R_L}{R_L} + \frac{C_{GS}C_L}{R_L} \\ C = g_m + \frac{1}{R_L} \end{cases} \\ \text{BJT: } \begin{cases} A = R_s r_\pi (C_\pi C_e + C_\pi C_L + C_e C_L) \\ B = g_m r_\pi R_s C_\pi + r_\pi (C_e + C_L) + R_s (C_e + C_L) + \frac{R_s r_\pi}{R_L} (C_e + C_L) \\ C = g_m r_\pi + \frac{R_s}{R_L} + \frac{r_\pi}{R_L} + 1 \end{cases} \end{cases}$$

# Chapter 12 Feedback

## 12.1 Ideal Feedback Network

In this chapter, we deal with the fundamentals of (negative) feedback and its application to electronic circuits, especially amplifiers.

$$A_{CL} = \frac{Y}{X} = \frac{A}{1 + KA}, \quad \text{Loop Gain} := KA$$

$$\begin{cases} A = A_0 \\ \omega_p = \omega_0 \\ \text{GBW} = A_0 \omega_0 \\ \omega_u = \sqrt{A_0^2 - 1} \omega_0 \end{cases} \mapsto \begin{cases} A = \frac{A_0}{1 + KA_0} \\ \omega_p = (1 + KA_0) \omega_0 \\ \text{GBW} \equiv A_0 \omega_0 \\ \omega_u = \sqrt{A_0^2 - (1 + KA_0)^2} \omega_0 \end{cases}$$

$$\text{Transfer function: } H(s) \mapsto \frac{H_0}{1 + KH_0}$$

$$\text{Vol amp.: } \begin{cases} Z_{in} = Z_{in0} \\ Z_{out} = Z_{out0} \end{cases} \mapsto \begin{cases} Z_{in} = (1 + KA_0) Z_{in0} \\ Z_{out} = \frac{Z_{out0}}{(1 + KA_0)} \end{cases}$$

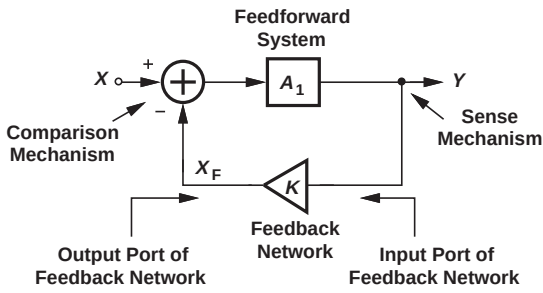


Figure 12.1: General feedback system

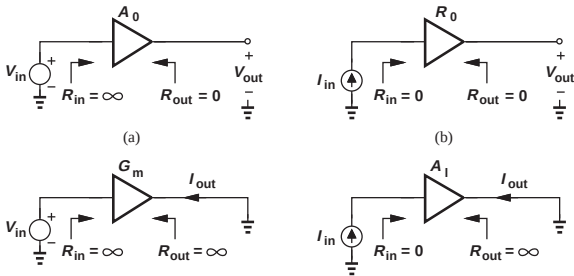


Figure 12.2: Types of amplifiers: (a) Voltage, (b) transimpedance, (c) transconductance, and (d) current amplifiers.

**Note the current direction of  $G_m$  and  $A_I$  amplifiers.**

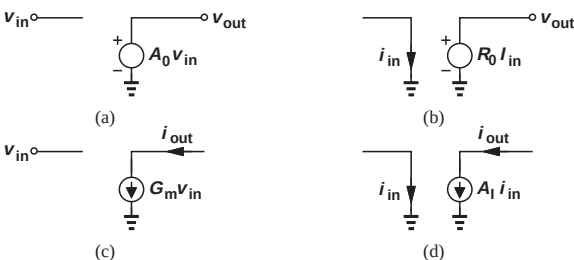


Figure 12.3: Ideal models for (a) voltage, (b) transimpedance, (c) transconductance, and (d) current amplifiers.

## 12.2 Nonideal Feedback Network

Breaking rules: “connected to another port of ideal forward system” or “full-sense but zero-return”.

- (1) Identify the forward amplifier and feedback network;
- (2) Break the feedback loop (according to the rules described below) and calculate open-loop parameters;
- (3) Determine the feedback factor  $K$ ;
- (4) Calculate the closed-loop parameters.

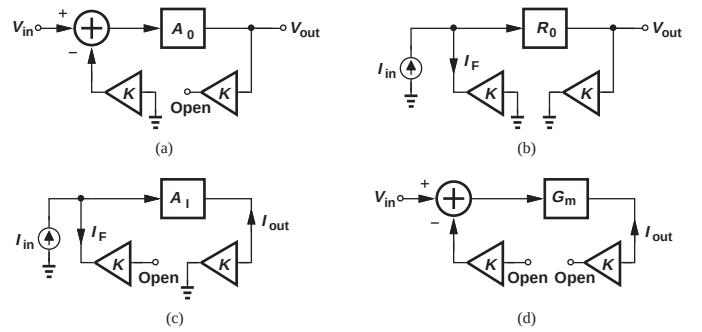


Figure 12.4: Proper termination of duplicates in feedback.

## 12.3 Stability and Compensation

A well-behaved system typically requires a phase margin of  $60^\circ$ , in other words, the gain crossover frequency  $\omega_{GX}$  must fall below the **second pole**.

$$\begin{aligned} \text{gain crossover freq. : } \omega_{GX} &= \omega |_{KH|=1} \\ \text{phase crossover freq. : } \omega_{PX} &= \omega |_{\angle KH = -180^\circ} \end{aligned} \quad (12.1)$$

## Chapter 13 Oscillators

To be completed.

# Chapter 14 Output Stages and Power Amplifiers

This chapter deals with circuits that can provide a high output power. We first reexamine circuits studied in previous chapters to understand their shortcomings for this task. Next, we introduce the “push-pull” stage and various modifications to improve its performance.

## 14.1 Emitter follower as power amp

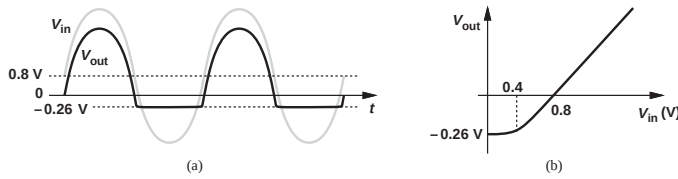


Figure 14.1: Emitter follower as power amplifier

## 14.2 Push-Pull Stage

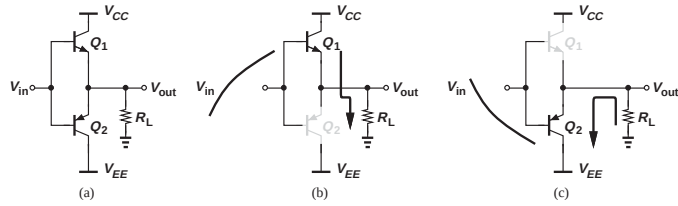


Figure 14.2: Basic push-pull stage

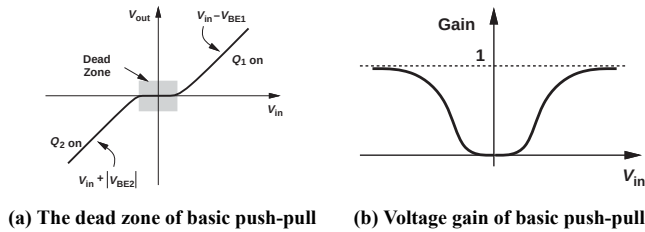


Figure 14.3: Basic push-pull stage characteristics

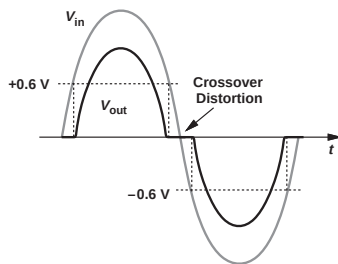


Figure 14.4: Input-output waveform of push-pull stage

## 14.3 Improved Push-Pull Stage

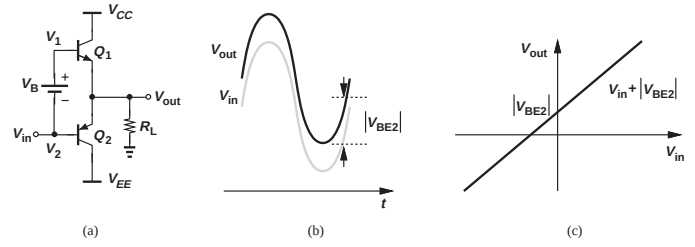


Figure 14.5: Improved push-pull stage

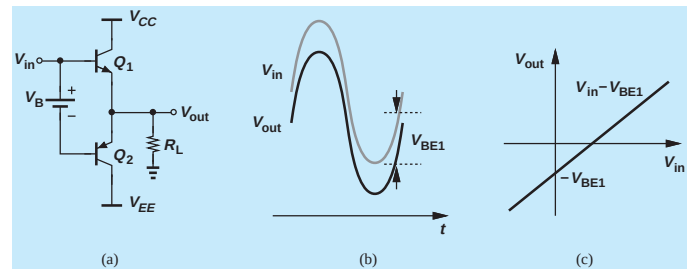


Figure 14.6: Improved push-pull stage with  $Q_1$  input

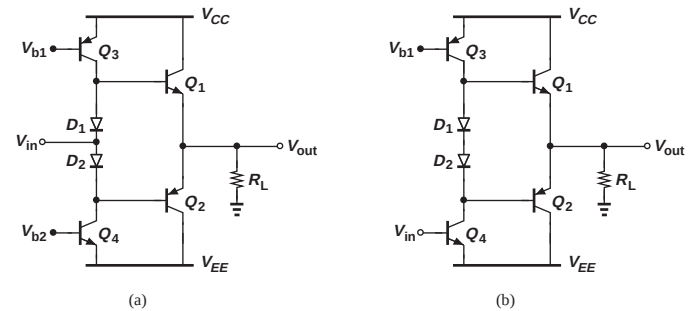


Figure 14.7: Addition of CE stage for push-pull stage

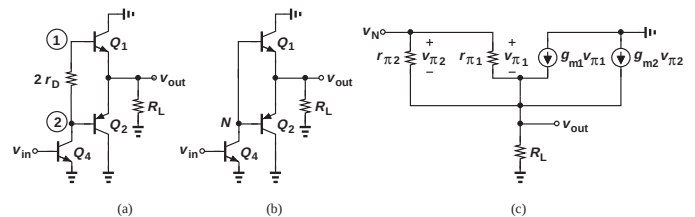


Figure 14.8: Overall voltage gain of improved push-pull stage

### 14.3.1 Small-Signal Analysis

Note that  $Q_1$  and  $Q_2$  are connected in parallel, because their collectors are tied to ac-ground and the emitters are tied together (if  $r_{D1}$  and  $r_{D2}$  is sufficiently small).

Therefore, in small-signal analysis<sup>①</sup>, we can combine  $Q_1$  and  $Q_2$  as a single transistor  $Q_1 + Q_2$  with the following

<sup>①</sup>See <https://yidingg.github.io/YiDingg/#/Blogs/Electronics/Improved%20Push-Pull%20Stage> for the derivations.

parameters:

$$g_m = g_{m1} + g_{m2} \quad (14.1)$$

$$r_\pi = r_{\pi1} \parallel r_{\pi2}, \quad r_o = r_{O1} \parallel r_{O2} \quad (14.2)$$

$$\beta = g_m r_\pi = (g_{m1} + g_{m2}) (r_{\pi1} \parallel r_{\pi2}) = \quad (14.3)$$

yielding the voltage gain and output impedance<sup>②</sup>:

$$R_N = r_\pi + (\beta + 1)(R_L \parallel r_O) \quad (14.4)$$

$$A_1 = -g_{m4} (R_N \parallel r_{O3} \parallel r_{O4}) \quad (14.5)$$

$$A_2 = g_m (R_L \parallel r_O) \quad (14.6)$$

$$A_v = A_1 A_2 \approx -g_{m4} [r_\pi + (\beta + 1)R_L] \cdot g_m R_L \quad (14.7)$$

$$R_{out} \approx \frac{1}{g_{m1} + g_{m2}} + \frac{r_{O3} \parallel r_{O4}}{(g_{m1} + g_{m2}) (r_{\pi1} \parallel r_{\pi2})} \quad (14.8)$$

### 14.3.2 Large-Signal Analysis

It is obvious that the push-pull stage operates in a large-signal mode so that the small-signal analysis is of limited value, meriting an analysis of the large-signal behavior.

### 14.3.3 Power Dissipation and Efficiency

We only consider the approximate power dissipation and efficiency of the improved push-pull stage, and they are given by:

$$P_{Q,av} = P_{S,av} - P_{O,av}$$

$$P_{S,av} = \frac{2V_P V_{CC}}{\pi R_L}, \quad P_{O,av} = \frac{V_P^2}{2R_L}$$

$$\eta = \frac{P_{O,av}}{P_{S,av}} = \frac{\pi V_P}{4 V_{CC}} < \frac{\pi}{4} \approx 78.5\% \quad (V_P = V_{CC})$$

$$P_{Q,peak} = \frac{V_{CC}^2}{\pi^2 R_L} \quad \left( V_P = \frac{2}{\pi} V_{CC} \approx V_{CC} \times 64\% \right)$$

<sup>②</sup> See [Yi Ding's Website > Blogs > Electronics > Improved Push-Pull Stage](https://yidingg.github.io/YiDingg/#/Blogs/Electronics/Improved%20Push-Pull%20Stage) (<https://yidingg.github.io/YiDingg/#/Blogs/Electronics/Improved%20Push-Pull%20Stage>)

## Chapter 15 Analog Filters

An overview of analog filters, another crucial component that is widely utilized in electronic systems, is given in this chapter.

### 15.1 Passive Filters

relative sensitivity:  $S_C^P = S_{Component}^{Parameter} := \frac{\frac{dP}{dC}}{\frac{dP}{dC}}$

general 2nd-order filters:  $H(s) = \frac{\alpha s^2 + \beta s + \gamma}{s^2 + \frac{\omega_n}{Q}s + \omega_n^2} = H_0 \cdot \frac{fs^2 + gs + 1}{\frac{s^2}{\omega_n^2} + \frac{s}{\omega_n Q} + 1}$

$$\Delta = \frac{\omega_n^2}{Q^2} - 4\omega_n^2 \leq 0 \implies Q \geq \frac{1}{2} \implies p_{1,2} = \omega_n \cdot \frac{-1 \pm j\sqrt{4Q^2 - 1}}{2Q}, |p_{1,2}| = \omega_n$$

low-pass:  $\left(\frac{\gamma}{\omega_n^2}\right), \omega_{peak} = \omega_n \sqrt{1 - \frac{1}{2Q^2}}$   $\left\{ \begin{array}{l} \text{peak occurs if } Q > \frac{\sqrt{2}}{2} = 0.707 \\ \text{normalized peaking magnitude:} \\ \frac{H_{peak}}{H_0} = \frac{Q}{\sqrt{1 + \frac{1}{4Q^2}}} = \frac{2Q^2}{\sqrt{4Q^2 - 1}} \end{array} \right.$

high-pass:  $\left(\frac{\alpha s^2}{\omega_n^2}\right), \omega_{peak} = \frac{\omega_n}{\sqrt{1 - \frac{1}{2Q^2}}}$

band-pass:  $\left\{ \begin{array}{l} \left(\frac{\beta s}{\omega_n}\right), \omega_{peak} = \omega_n, H(\omega_n) = \frac{\beta}{\omega_n Q} = \frac{\beta Q}{\omega_n} \\ \omega_{3dB} = \omega_n \left[ \sqrt{1 + \frac{1}{4Q^2}} \pm \frac{1}{2Q} \right] = \frac{\omega_n}{2Q} \left[ \sqrt{4Q^2 + 1} \pm 1 \right] \\ BW = \omega_2 - \omega_1 = \frac{\omega_n}{Q}, \text{ normalized BW} = \frac{BW}{\omega_n} = \frac{1}{Q} \end{array} \right.$

### 15.2 Active Filters

$$H(s) = \frac{\gamma}{\frac{s^2}{\omega_n^2} + \frac{s}{\omega_n Q} + 1} = \frac{1}{s^2(R_2 R_2 C_1 C_2) + s(R_1 + R_2)C_2 + 1} \quad (15.1)$$

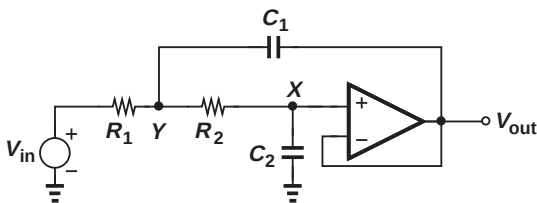
$$\omega_n = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}, \quad Q = \frac{1}{b\omega_n} = \frac{\sqrt{R_1 R_2 C_1 C_2}}{(R_1 + R_2)C_2} \quad (15.2)$$

$$H(s) = \frac{1}{s^2 R_1 R_2 C_1 C_2 + s[(R_1 + R_2)C_2 - (A_v - 1)R_1 C_1] + 1} \quad (15.3)$$

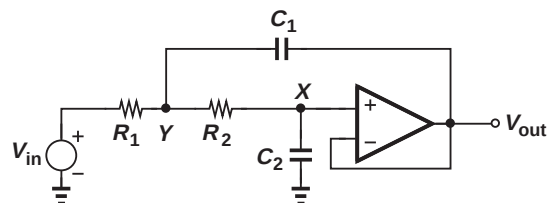
$$\omega_n = \frac{1}{\sqrt{R_1 R_2 C_1 C_2}}, \quad Q = \frac{\sqrt{R_1 R_2 C_1 C_2}}{(R_1 + R_2)C_2 - (A_v - 1)R_1 C_1} \quad (15.4)$$

with  $R_1 = R_2 = R, C_1 = C_2 = C$ , we have:  $\omega_n = \frac{1}{RC}, Q = \frac{1}{3 - A_v}$  (15.5)

sensitivity:  $\left\{ \begin{array}{l} S_{R_1}^{\omega_n} = S_{R_2}^{\omega_n} = S_{C_1}^{\omega_n} = S_{C_2}^{\omega_n} = -\frac{1}{2} \\ S_{R_1}^Q = -S_{R_2}^Q = -\frac{1}{2} + Q\sqrt{\frac{R_2 C_2}{R_1 C_1}} \\ S_{C_1}^Q = -S_{C_2}^Q = -\frac{1}{2} + Q\left(\sqrt{\frac{R_2 C_2}{R_1 C_1}} + \sqrt{\frac{R_1 C_1}{R_2 C_2}}\right) \\ S_{A_v}^Q = \frac{A_v Q}{\sqrt{\frac{R_2 C_2}{R_1 C_1}}} \end{array} \right. \quad (15.6)$



(a) Basic Sallen and Key filters



(b) Sallen and Key filters with in-band gain

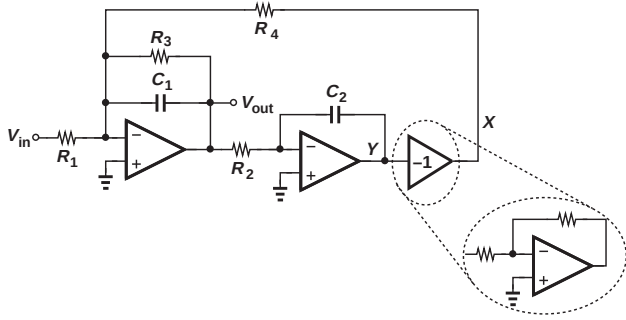
Figure 15.1: Sallen and Key filters (SKF)

## 15.3 Integrator-Based Biquads

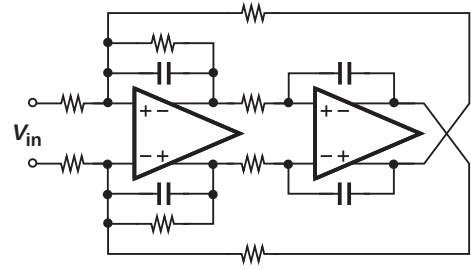
“Biquad” refers to a second-order filter with two poles and two zeros. It can be proved that the sensitivities of the GIC and the resulting filters with respect to component values are equal to 0.5 or 1.

$$\text{band-pass: } \frac{V_{out}}{V_{in}}(s) = -\frac{R_2 R_3 R_4}{R_1} \cdot \frac{s}{s^2 R_2 R_3 R_4 C_1 + s R_2 R_4 + \frac{R_3}{C_2}} = \frac{-\frac{s}{R_1 C_1}}{s^2 + \frac{s}{R_3 C_1} + \frac{1}{R_2 R_4 C_1 C_2}} \quad (15.7)$$

$$\text{low-pass: } \frac{V_{out}}{-s R_2 C_2} = V_Y \Rightarrow \frac{V_Y}{V_{in}}(s) = \frac{R_3 R_4}{R_1 C_2} \cdot \frac{1}{s^2 R_2 R_3 R_4 C_1 + s R_2 R_4 + \frac{R_3}{C_2}} = \frac{\frac{1}{R_1 R_2 C_1 C_2}}{s^2 + \frac{s}{R_3 C_1} + \frac{1}{R_2 R_4 C_1 C_2}} \quad (15.8)$$



(a) Tow-Thomas biquad



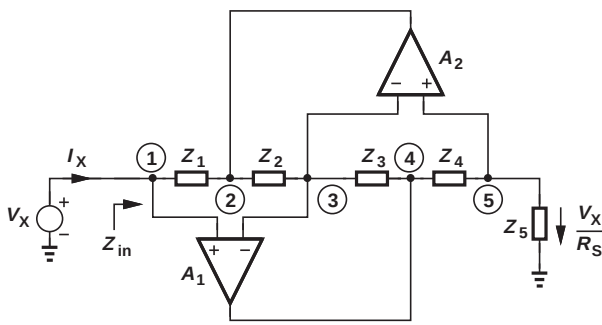
(b) Differential Tow-Thomas biquad

Figure 15.2: Tow-Thomas biquad and its differential form

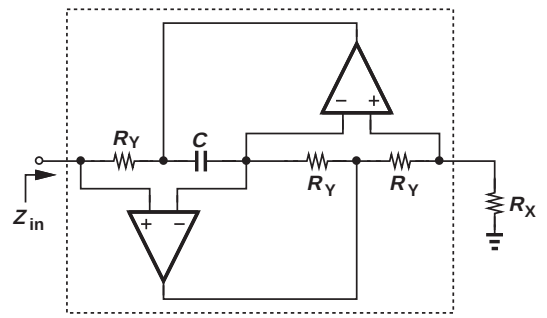
## 15.4 Impedance Converters

Note that a floating (two-terminals) impedance can not be achieved by canceling the GND node of  $Z_5$ . It can be proved that the sensitivities of the GIC and the resulting filters with respect to component values are equal to 0.5 or 1, i.e.,  $S_{R_1, R_2, R_3, R_4}^Q = 0.5$  and Such circuits therefore prove useful in both discrete and integrated applications.

$$Z_{in} = \frac{Z_1 Z_3}{Z_2 Z_4} Z_5, \quad Z_1 = Z_3 = Z_4 = R_Y, \quad Z_5 = R_X, \quad Z_2 = \frac{1}{sC} \Rightarrow Z_{in} = s R_X R_Y C, \quad L_{eq} = R_X R_Y C \quad (15.9)$$



(a) general impedance converter



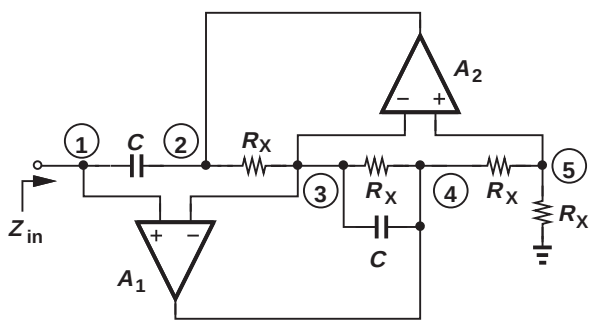
(b) emulated inductor

Figure 15.3: General impedance converter (GIC) emulating an inductor

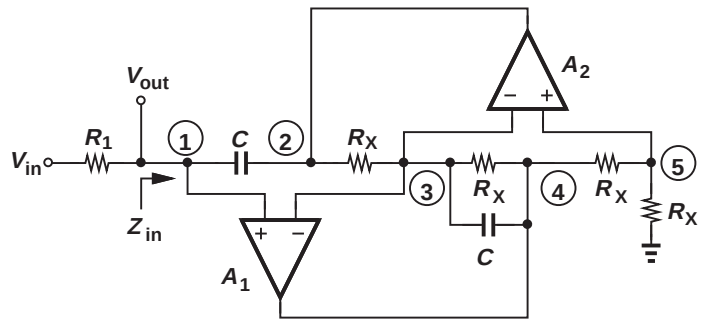
$$Z_{cap} = \frac{1}{sC(1 + sR_X C)}, \quad H(s) = \frac{V_{out}}{V_{in}} = \frac{1}{s^2 R_1 R_2 C^2 + s R_1 C + 1}, \quad \omega_n = \frac{1}{\sqrt{R_1 R_2 C^2}}, \quad Q = \frac{1}{b \omega_n} = \sqrt{\frac{R_2}{R_1}} \quad (15.10)$$

$$V_4 = \left(1 + \frac{R_Y}{R_X}\right) V_{out} = 2 V_{out} \Rightarrow H(s) = \frac{V_4}{V_{in}} = \frac{2}{s^2 R_1 R_2 C^2 + s R_1 C + 1}, \quad \omega_n = \frac{1}{\sqrt{R_1 R_2 C^2}}, \quad Q = \sqrt{\frac{R_2}{R_1}} \quad (15.11)$$





(a) Super capacitor using GIC



(b) 2nd-order low-pass filter using the super capacitor

Figure 15.4: Super capacitor using the general impedance converter

## Chapter 16 Digital CMOS Circuits

To be completed.

# Reference

- [1] Behzad Razavi. *Fundamentals of Microelectronics*. University of California Press, 2nd edition, 2014.
- [2] Behzad Razavi. *Design of Analog CMOS Integrated Circuits*. McGraw-Hill Education, 2nd edition, 2017.